



AKTU DSTL



DISCRETE MATHEMATICS

ONE SHOT REVISION

UNIT-II

FUNCTIONS & BOOLEAN ALGEBRA



LIVE

17 MARCH 9:00PM

UNIT -2 : AKTU Syllabus

Functions

- Definition of Function
- Classification of functions
- Operations on functions
- Growth of function

Boolean Algebra:

- Introduction
- Axioms and Theorems of Boolean Algebra
- Algebraic manipulation of Boolean Expressions
- Simplification of Boolean functions
- Karnaugh maps

Gateway Classes : 7455 9612 84

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : Functions and Boolean Algebra

Lecture - 01

Today's Target

- ✓ ➤ *Definition of Function*
- ✓ ➤ *Classification of functions*
- PYQ
- DPP

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$$R: A \rightarrow B$$

Function

$$R \subseteq A \times B$$

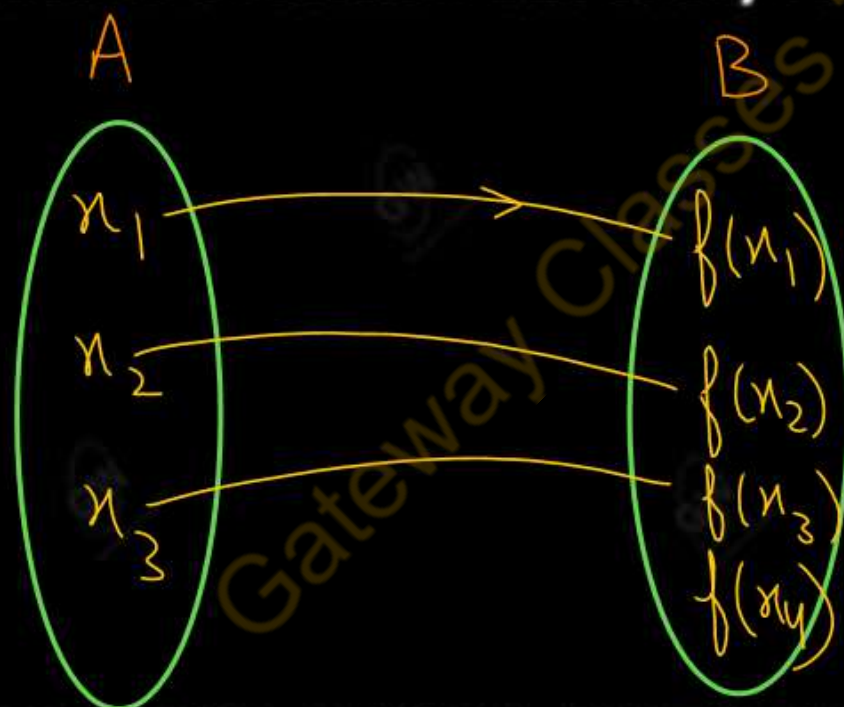
Let A and B are two non-empty sets

A function $f: A \rightarrow B$ is a relation / Mapping from set A to set B , such that each and every element of Set A is related to exactly one element of set B .

OR

Those relations in which each and every element of set A has unique image in Set B

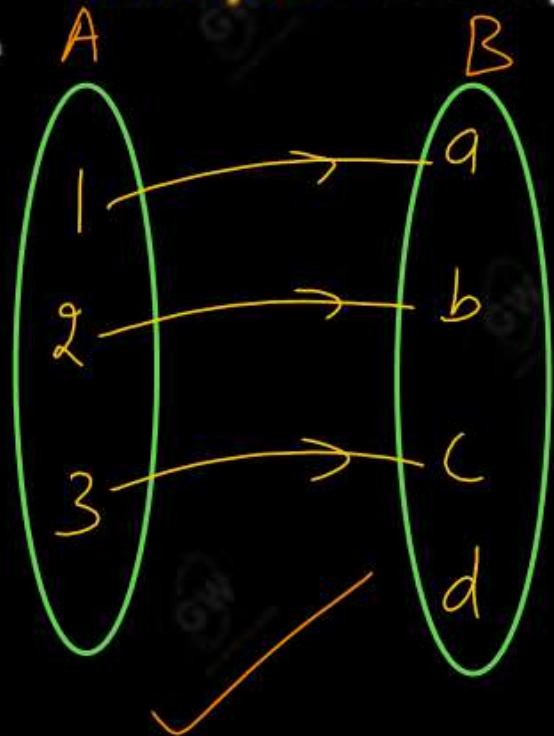
Conclusion : Every function is a relation but every relation is not a function .



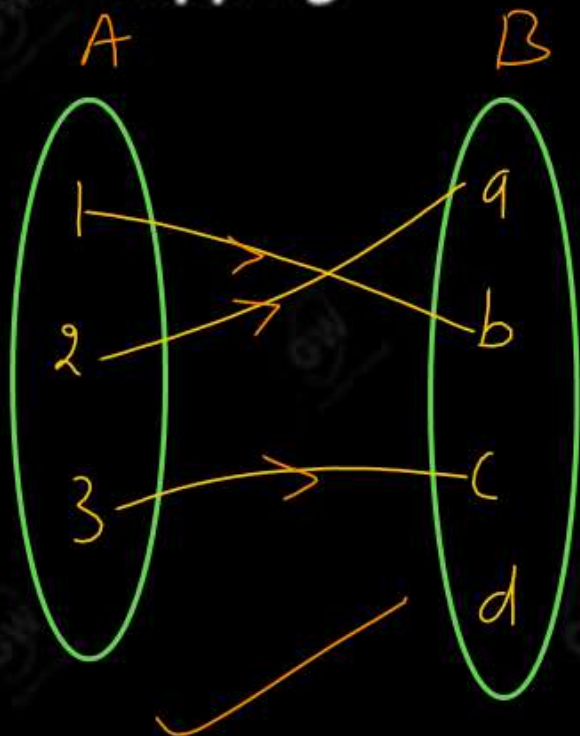
Note: $f(x_1)$ is called image of x_1 while x_1 is called pre-image of $f(x_1)$

Example : Identify Function Mapping

(i)



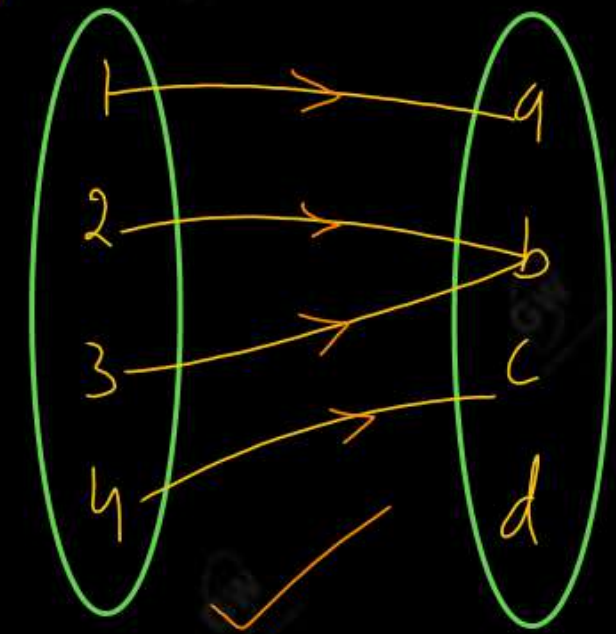
(ii)



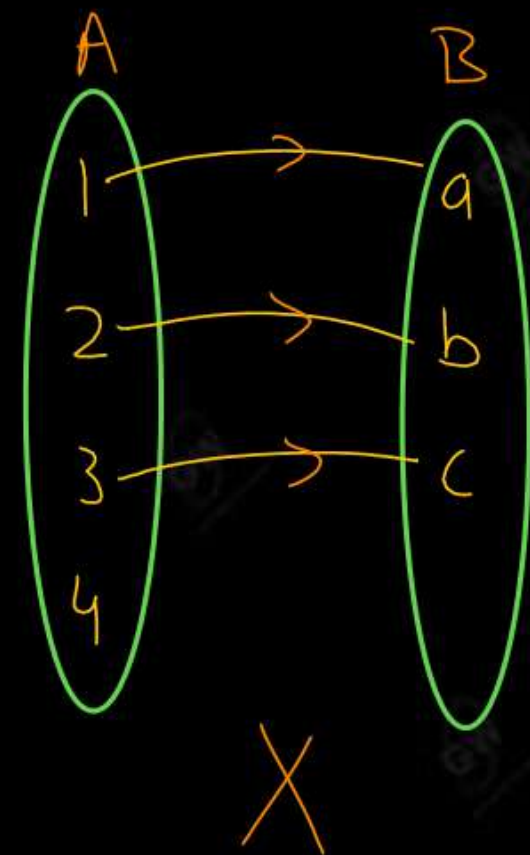
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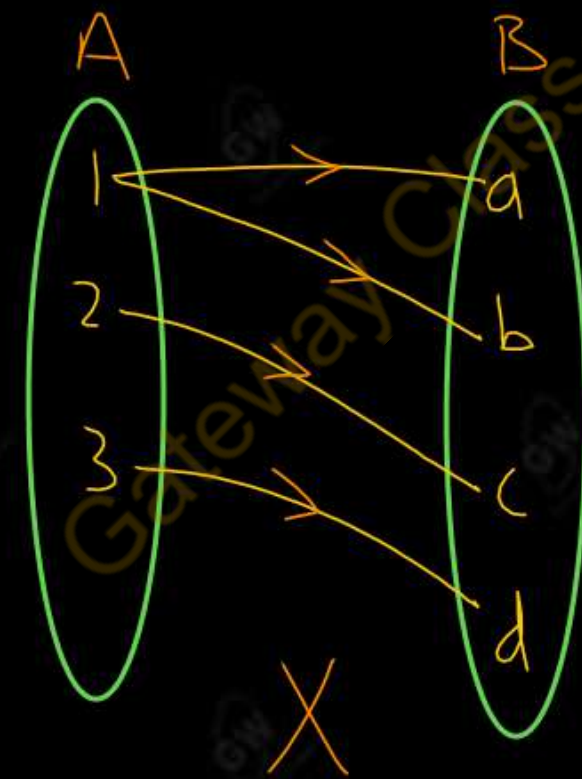
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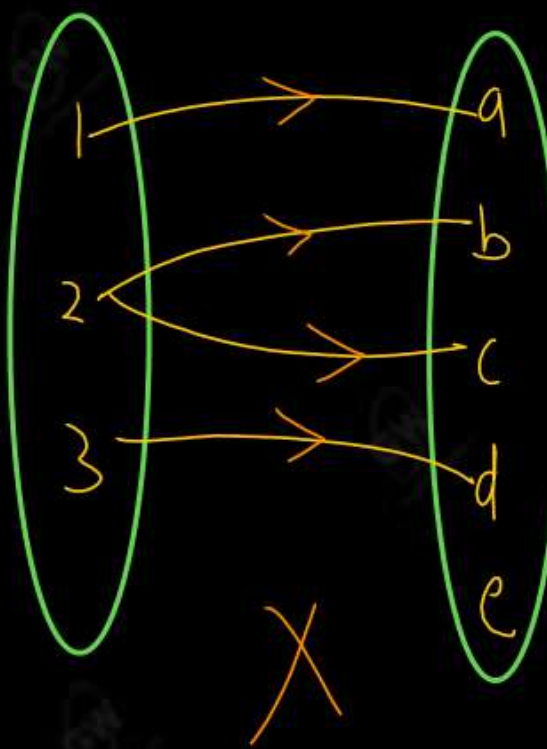
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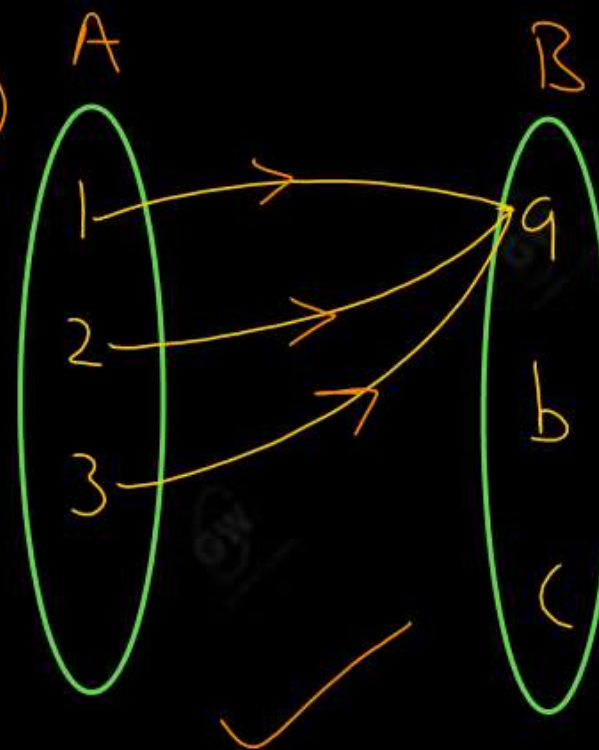
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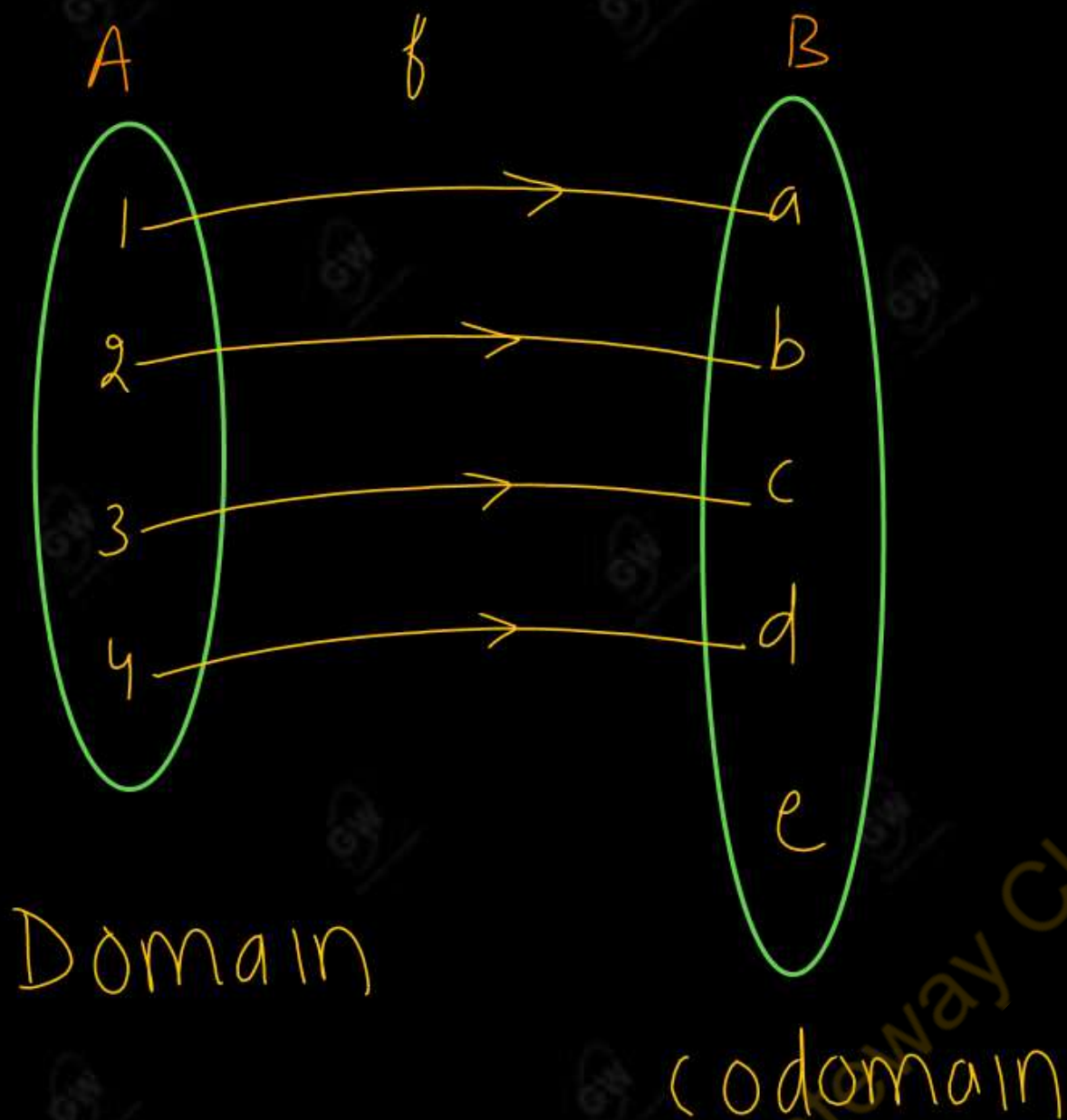
(vii)



(viii)



Domain, Co-domain and Range of function



$$\text{Dom}(f) = \{1, 2, 3, 4\}$$

$$\text{codomain} = \{a, b, c, d, e\}$$

$$\text{Range}(f) = \{a, b, c, d\}$$

$$\begin{array}{ccc} f: & R & \longrightarrow R \\ & \downarrow & \downarrow \\ & \text{Domain} & \text{codomain} \end{array}$$

Note:- If $n(A) = m$ and $n(B) = n$, then total number of possible function from A to B

$$\text{number of functions} = (n)^m \quad (5)^4 =$$

Classifications of function (Mapping)

✓ (1) ONE –ONE FUNCTION (or injective function)

✓ (2) MANY-ONE FUNCTION

✓ (3) ONTO FUNCTION (or surjective function)

✓ (4) INTO FUNCTION

imp ✓ (5) ONE-ONE ONTO FUNCTION (Bijective function)

✓ (6) One-one into function

✓ (7) Many-one onto function

✓ (8) Many-one into function

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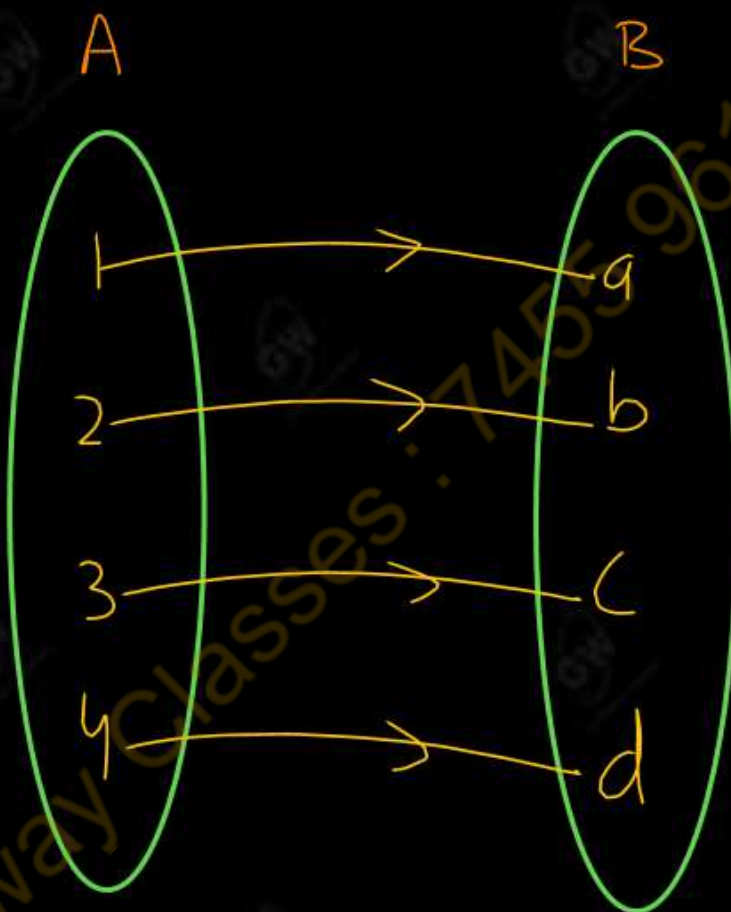
ONE – ONE FUNCTION OR One to one function OR Injective function

A function $f : A \rightarrow B$ is said to be one-one function if different elements of A have different images in B.

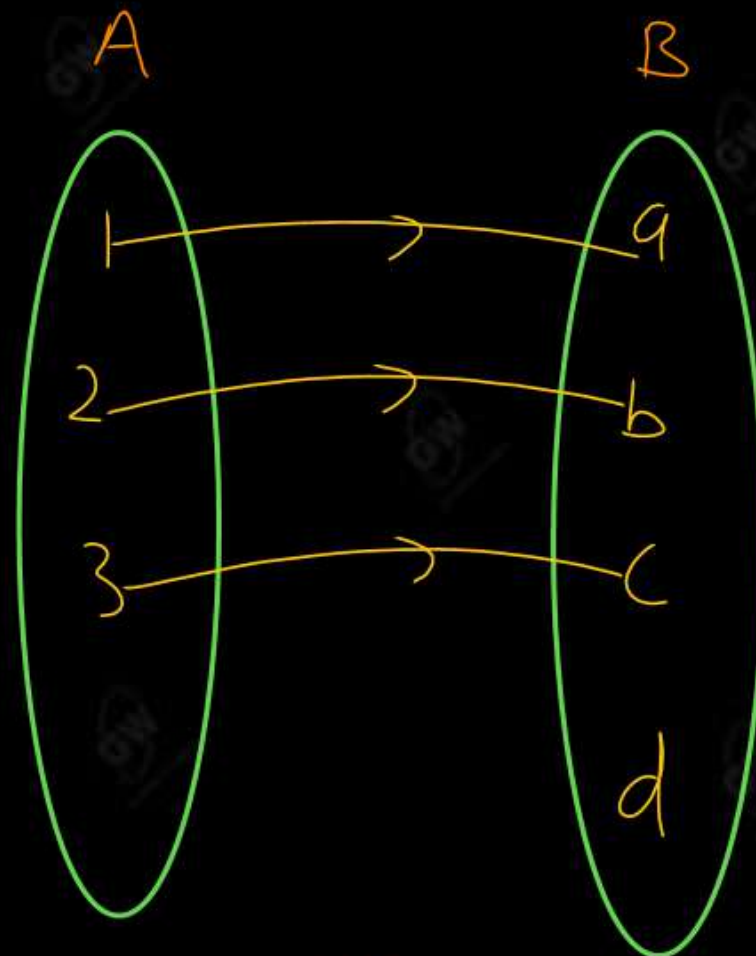
Thus, $f : A \rightarrow B$ is one-one if and only if

$$f(x_1) = f(x_2)$$

$$\Rightarrow \boxed{x_1 = x_2} \quad \forall x_1, x_2 \in A$$



(i)



(ii)

Note:- If $n(A) = m$ and $n(B) = n$, then

Total number of one-one function = ${}^n P_m = \frac{n!}{(n-m)!}$

Q. 1 :- Show that the function $f: N \rightarrow N$, given by $f(x) = 2x + 3$ is one - one

$$f(x) = 2x + 3$$

$$\text{Let } x_1, x_2 \in N$$

$$\text{Put } f(x_1) = f(x_2)$$

$$2x_1 + 3 = 2x_2 + 3$$

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

Hence, $f(x)$ is a one-one function

Q.2:- Check whether the function $f(x) = x^2 - 1$ is injective or not for $f: \mathbb{R} \rightarrow \mathbb{R}$

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$$f(x) = x^2 - 1$$

$$\text{Let } x_1, x_2 \in \mathbb{R}$$

$$\text{Put } f(x_1) = f(x_2)$$

$$x_1^2 - 1 = x_2^2 - 1$$

$$x_1^2 = x_2^2$$

$$x_1^2 - x_2^2 = 0$$

$$(x_1 - x_2)(x_1 + x_2) = 0$$

$$x_1 = x_2 \text{ or } x_1 = -x_2$$

Hence, $f(x)$ is not injective

Q.3:- Check whether the function $f(x) = x^3 + 2$ is one – one or not for $f: \mathbb{R} \rightarrow \mathbb{R}$.

$$f(x) = x^3 + 2$$

$$\text{Let } x_1, x_2 \in \mathbb{R}$$

$$\text{Put } f(x_1) = f(x_2)$$

$$x_1^3 + \cancel{2} = x_2^3 + \cancel{2}$$

$$x_1^3 = x_2^3$$

$$x_1 = x_2$$

Hence, $f(x)$ is one-one function

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MANY – ONE FUNCTION

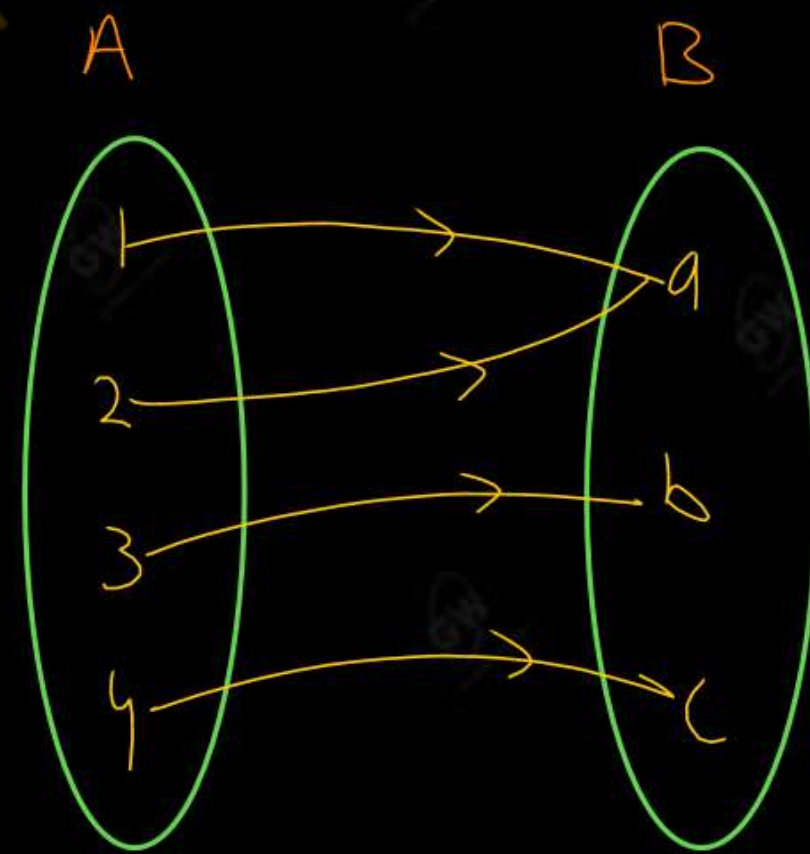
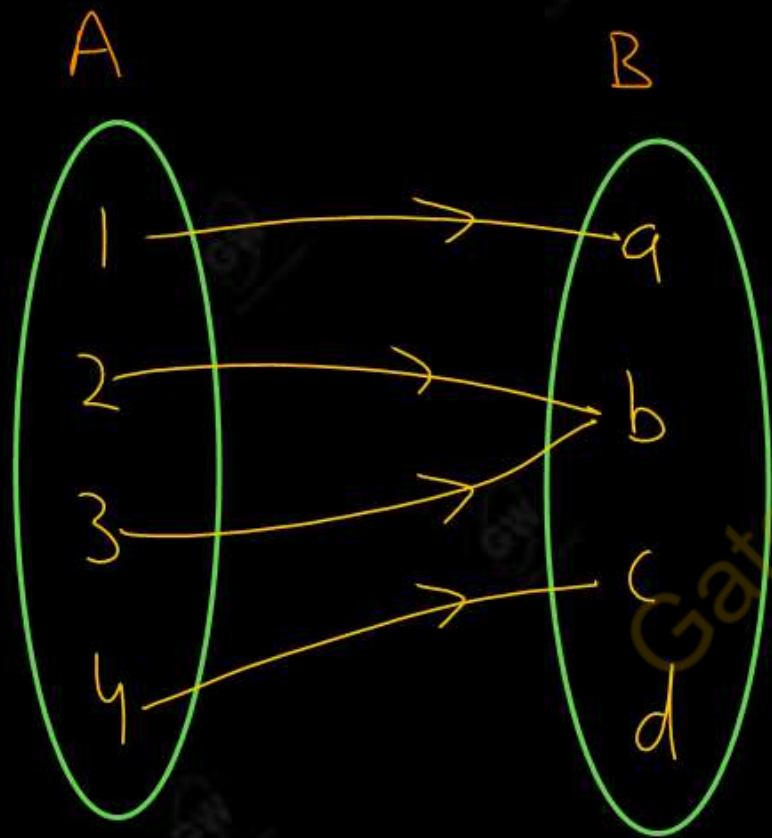
A function $f: A \rightarrow B$ is said to be a many – one function if two or more elements of set A have same image in B.

Thus, $f: A \rightarrow B$ is Many one if and only if

$$f(x_1) = f(x_2)$$

$$\Rightarrow x_1 \neq x_2 \quad \forall x_1, x_2 \in A$$

In other words, $f: A \rightarrow B$ is Many one if it is not one – one function

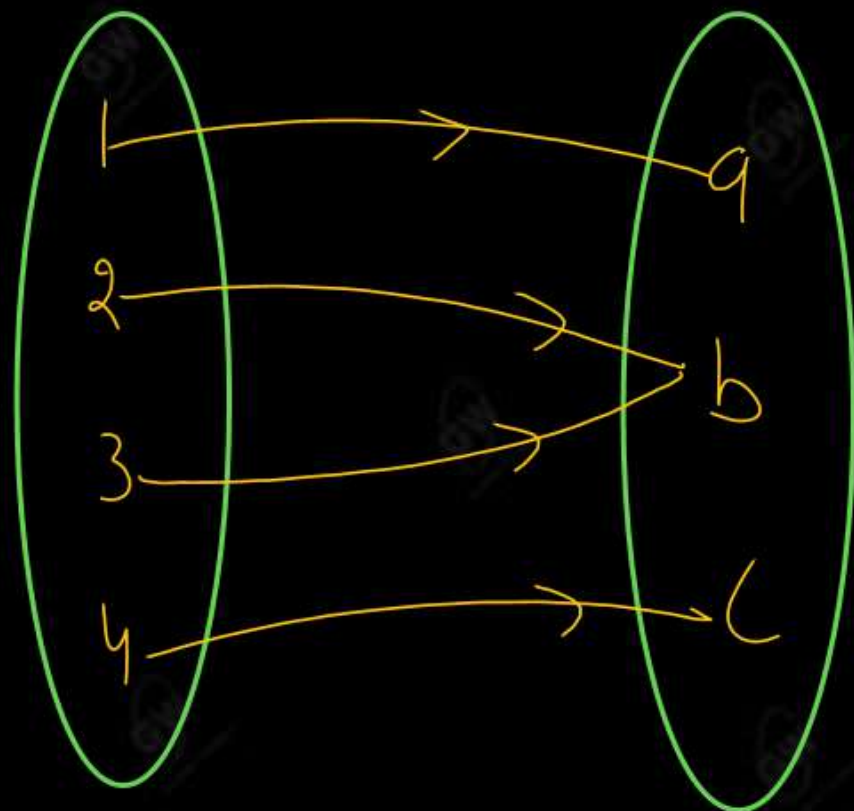
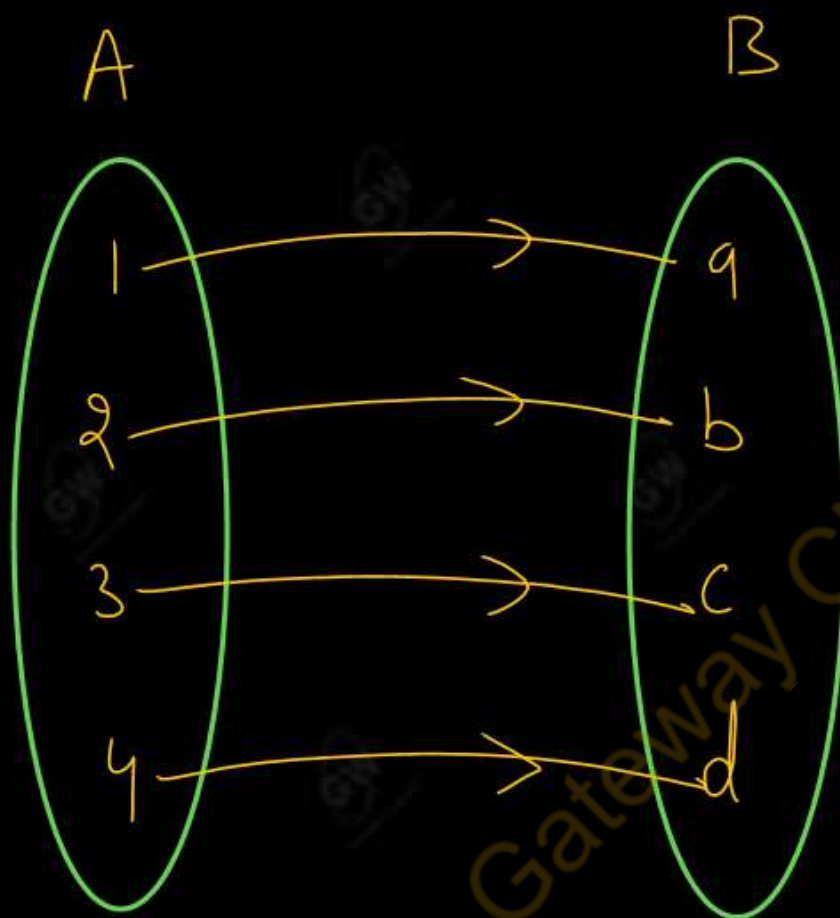


ONTO FUNCTION (or Surjective function)

A function $f: A \rightarrow B$ is said to be an ONTO function if every element of set B have pre-image in set A

i.e. Range of function = codomain of function

Example



Q.4:- Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}^+$ defined by $f(x) = x^2 \quad \forall x \in \mathbb{R}$ is an Onto function

$$f(x) = x^2$$

$$y = x^2$$

$$x^2 = y$$

$$x = \sqrt{y}$$

$$\Rightarrow y \in \mathbb{R}^+$$

Hence

Range of function = codomain of function

$\Rightarrow f(x)$ is an ONTO Function

Q.5:- Discuss the Surjectivity of the following functions

(i) $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3 + 2$ for $x \in \mathbb{R}$

(ii) $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2 + 2$ for $x \in \mathbb{R}$

(i) $f(x) = x^3 + 2$

$$y = x^3 + 2$$

$$y - 2 = x^3$$

$$x^3 = y - 2$$

$$x = (y - 2)^{1/3}$$

$$\Rightarrow y \in \mathbb{R}$$

Range of f = codomain of f

Hence

$f(x)$ is a surjective function

(ii) $f(x) = x^2 + 2$

$$y = x^2 + 2$$

$$x^2 = y - 2$$

$$x = \sqrt{y - 2}$$

$$\Rightarrow y \geq 2$$

Range of $f \neq$ codomain of f

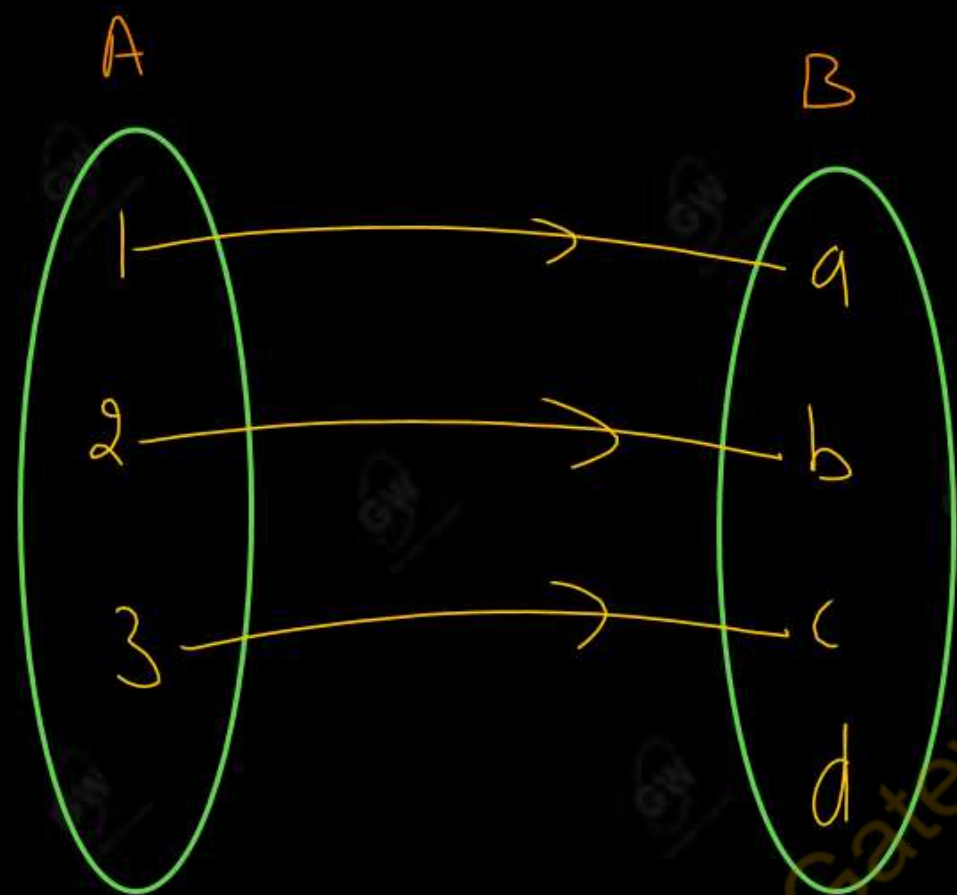
Hence, function is not surjective

INTO FUNCTION :

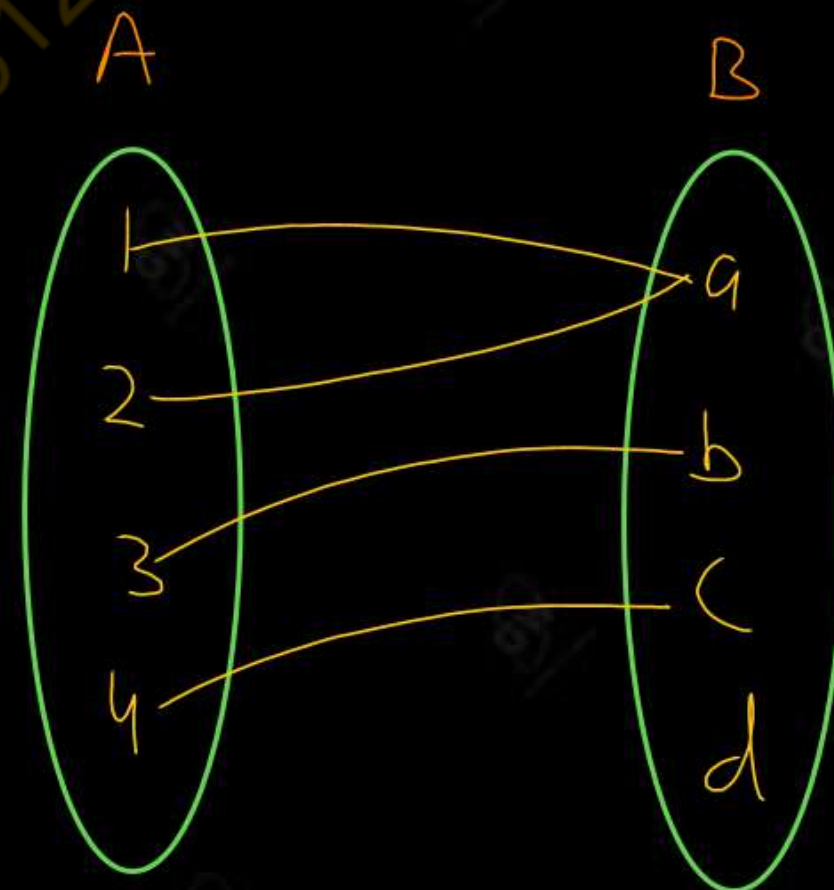
A function $f : A \rightarrow B$ is an Into function if there exist an element in B having no pre-image in A .

Thus, $f : A \rightarrow B$ is an Into function if it is not Onto function

Example



(i)



(ii)

ONE-ONE ONTO FUNCTION (Bijective function)

A function $f : A \rightarrow B$ is Bijective if it is

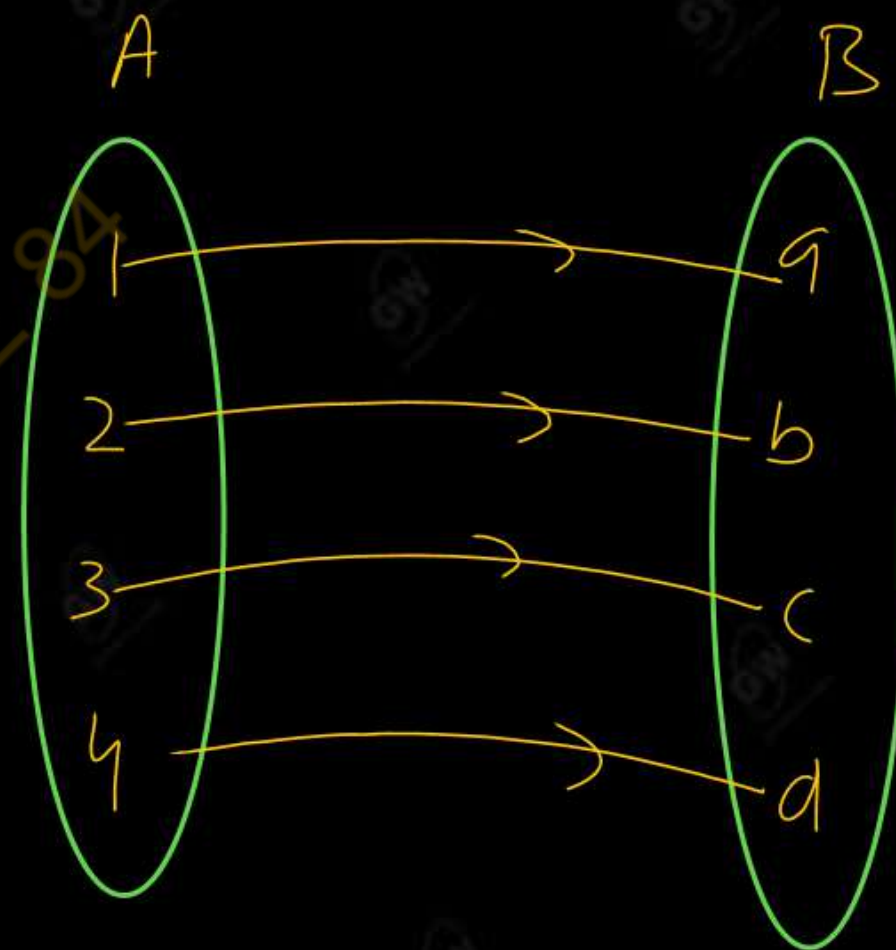
(i) **One – One Function**

$$f(x_1) = f(x_2)$$

$$\Rightarrow x_1 = x_2$$

(ii) **Onto Function**

For all $y \in B$, there exist $x \in A$ such that $f(x) = y$



Q.6:- Check the Injectivity or Surjectivity of the following function

(i) $f: N \rightarrow N$ given by $f(x) = x^2$

(ii) $f: Z \rightarrow Z$ given by $f(x) = x^2$

HW

(i) $f(n) = n^2$

Let $n_1, n_2 \in N$

Put $f(n_1) = f(n_2)$

$$n_1^2 = n_2^2$$

$$n_1 = n_2$$

Hence

$f(n)$ is injective

Again

$$f(n) = n^2$$

$$y = n^2$$

$$n^2 = y$$

$$n = \sqrt{y}$$

$$\text{Range}(f) \neq \text{codomain}(f)$$

$f(n)$ is surjective

Q.7:- Show that the function $f : \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{1\}$ given by $f(x) = \frac{x-2}{x-3}$ is a bijection

$$f(x) = \frac{x-2}{x-3}$$

Let $x_1, x_2 \in \mathbb{R} - \{3\}$

Put $f(x_1) = f(x_2)$

$$\frac{x_1-2}{x_1-3} = \frac{x_2-2}{x_2-3}$$

$$(x_1-2)(x_2-3) = (x_1-3)(x_2-2)$$

$$\cancel{x_1 x_2} - 3x_1 - 2x_2 + 6 = \cancel{x_1 x_2} - 2x_1 - 3x_2 + 6$$

$$-3x_1 - 2x_2 = -2x_1 - 3x_2$$

$$+x_1 = +x_2$$

$$x_1 = x_2$$

$f(x)$ is injective

Again

$$f(x) = \frac{x-2}{x-3}$$

$$y = \frac{x-2}{x-3}$$

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$$y(x-3) = x-2$$

$$xy - 3y = x - 2$$

$$xy - x = 3y - 2$$

$$x(y-1) = 3y-2$$

$$x = \frac{3y-2}{y-1}$$

$\Rightarrow$ Range = $\mathbb{R} - \{1\}$

$$\Rightarrow \text{Range}(f) = \text{codomain } f$$

Hence

$f(x)$ is surjective

Since $f(x)$ is injective as well as surjective

$\Rightarrow f(x)$ is bijective

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Topic : Classification of function

Q.1. Show that the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^2 + 1$ for all $x \in \mathbb{Z}$ is a many – one function.

Q.2. Show that $f : \mathbb{R} \rightarrow \mathbb{R}$, given by $f(x) = ax + b$ where $a, b \in \mathbb{R}, a \neq 0$ is a bijection

Q.3. Show that $f : \mathbb{R} \rightarrow \mathbb{R}$, defined as $f(x) = x^3$ is a bijection.

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Link in Description

Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS

(Discrete Mathematics)

UNIT - 2: Functions and Boolean Algebra

Lecture - 02

Today's Target

- *Inverse of a Function*
- PYQ
- DPP

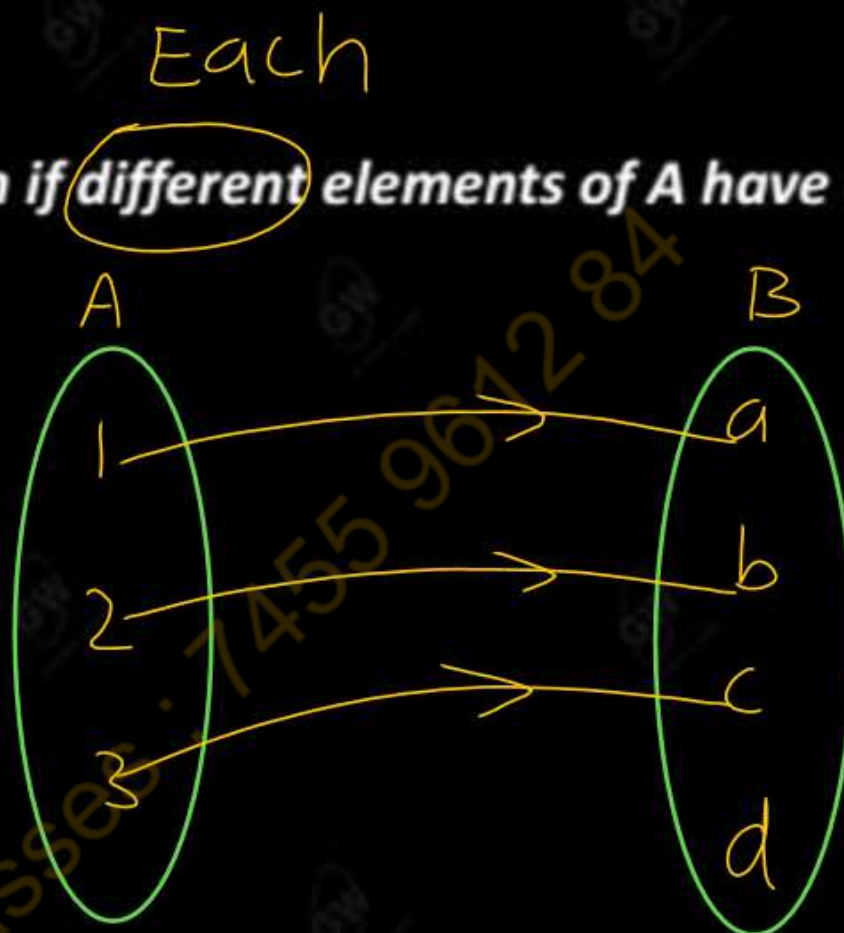
ONE - ONE FUNCTION

A function $f : A \rightarrow B$ is said to be one-one function if different elements of A have different images in B .

Thus, $f : A \rightarrow B$ is one-one if and only if

$$f(x_1) = f(x_2)$$

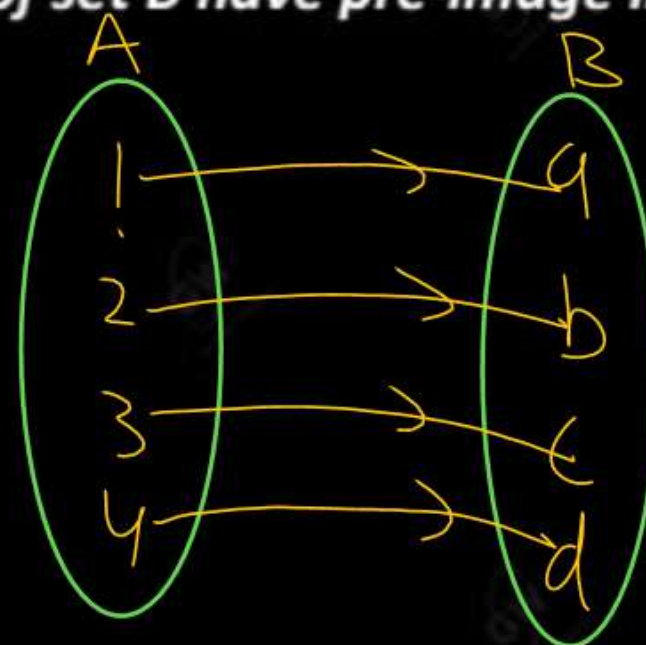
$$\Rightarrow x_1 = x_2 \quad \forall x_1, x_2 \in A$$



ONTO FUNCTION (or Surjective function)

A function $f: A \rightarrow B$ is said to be an ONTO function if every element of set B have pre-image in set A

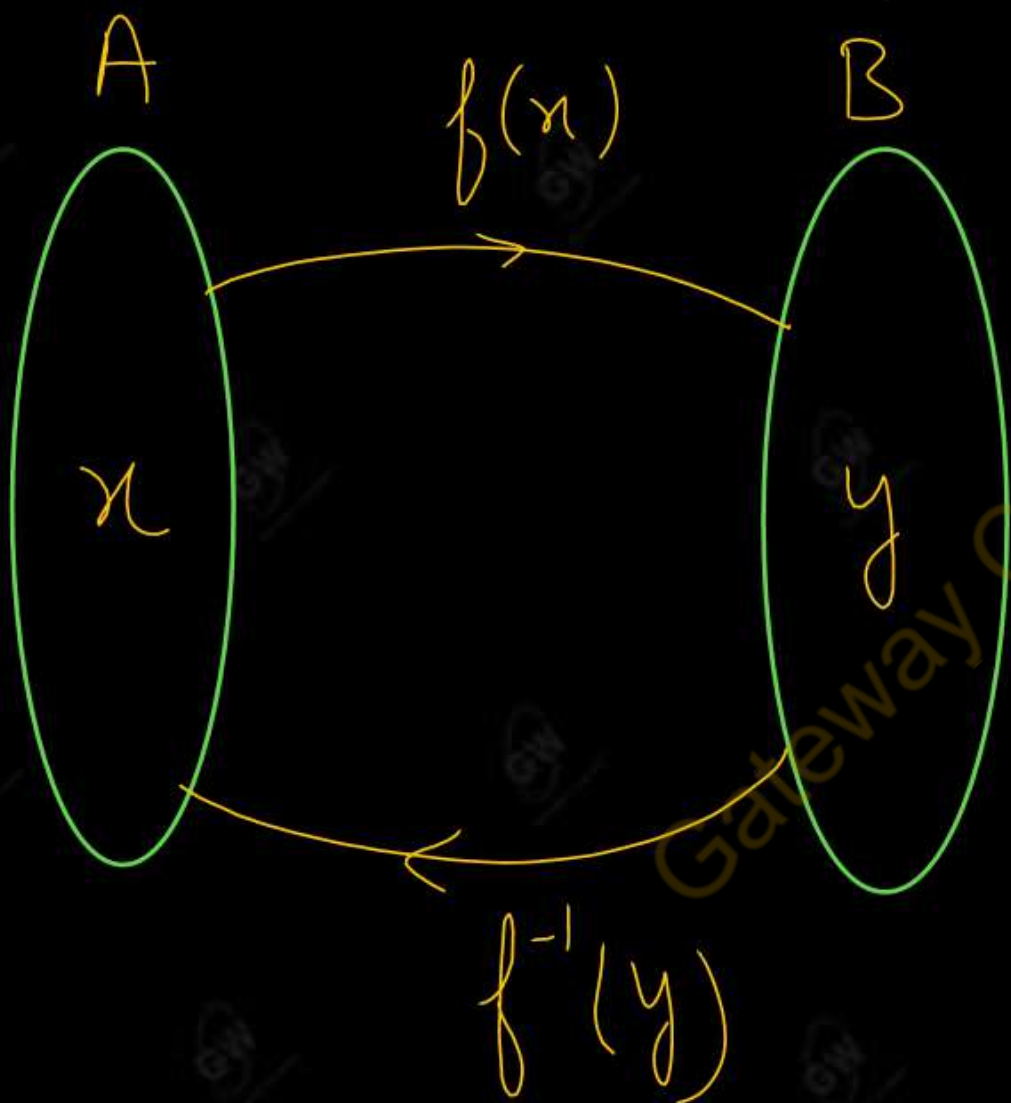
i.e. Range of function = codomain of function



Inverse of a Function

A function $f : A \rightarrow B$ is called invertible function if it is one one onto function and its inverse is given as

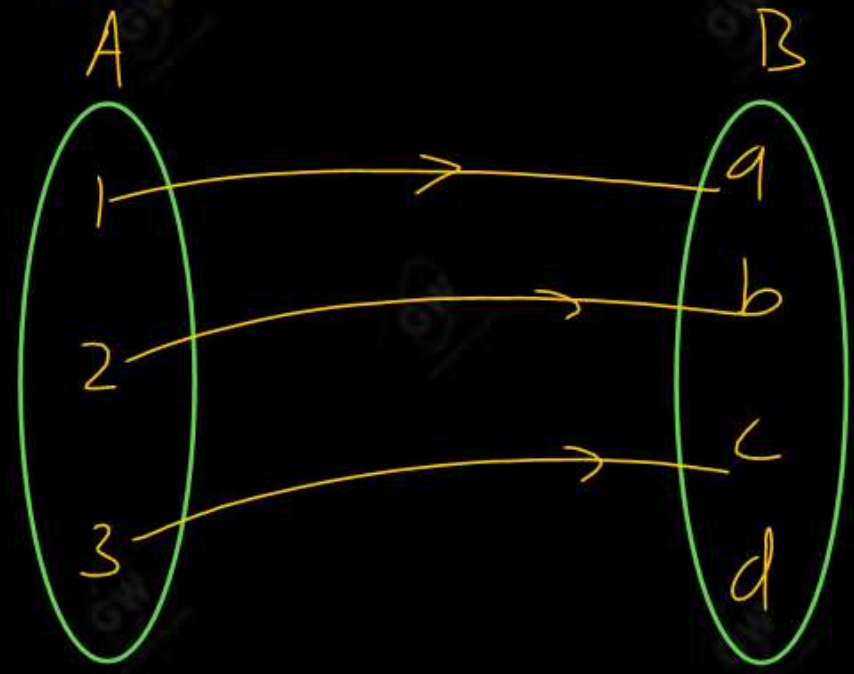
$$f^{-1} : B \rightarrow A$$



If $f(x) = y$ then

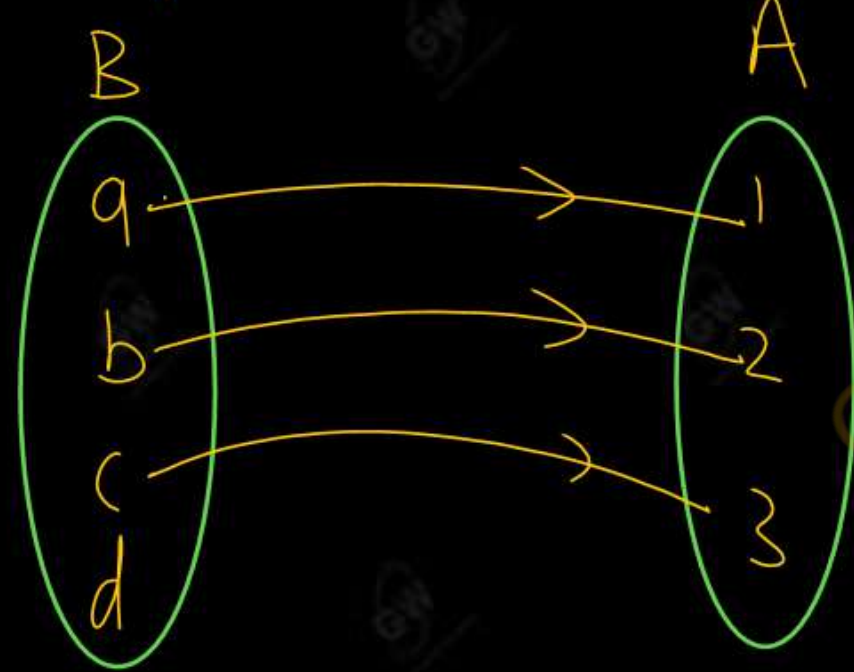
$$f^{-1}(y) = x$$

Example $f: A \rightarrow B$



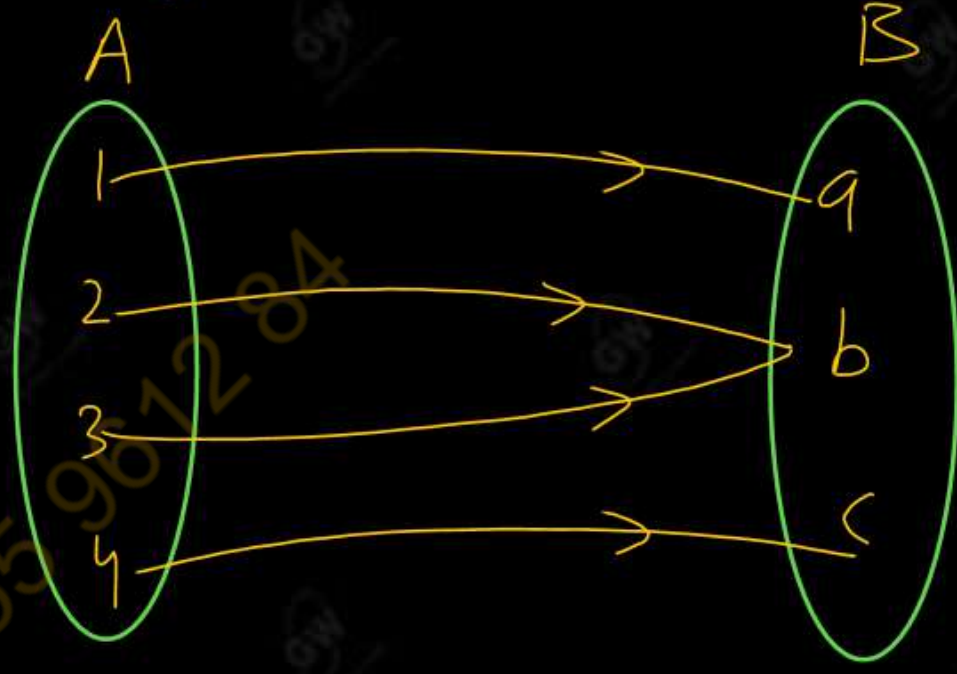
one-one
into

$f^{-1}: B \rightarrow A$ X



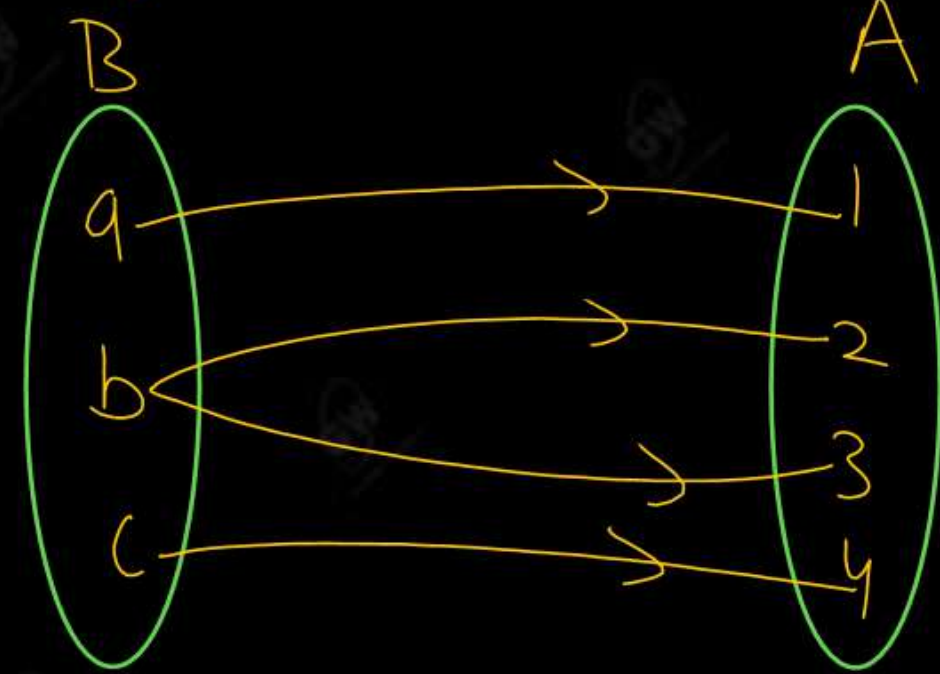
It is not
a function

$f: A \rightarrow B$



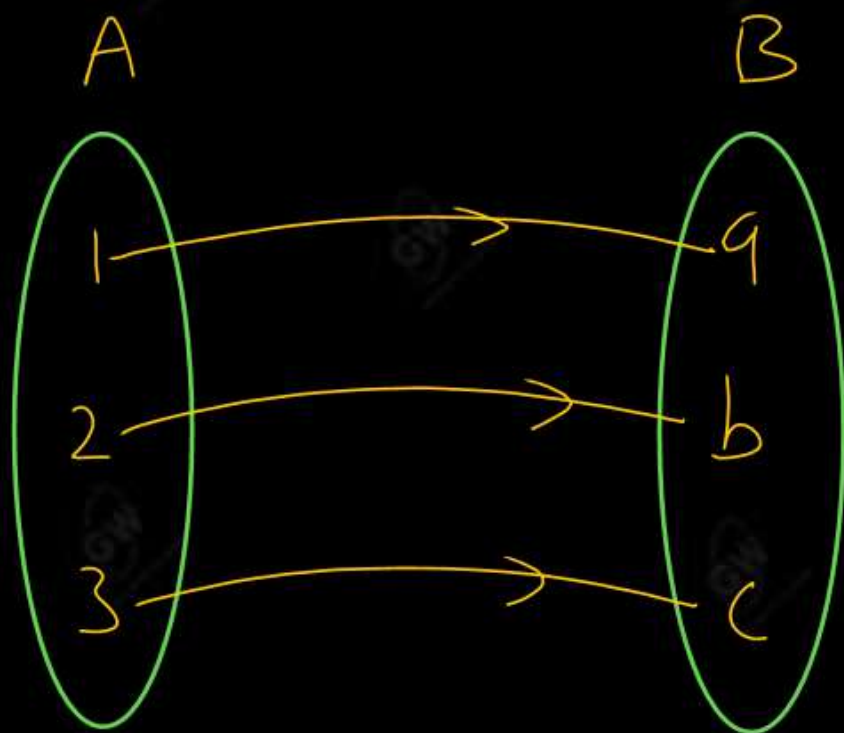
Many-one
onto

$f^{-1}: B \rightarrow A$ X



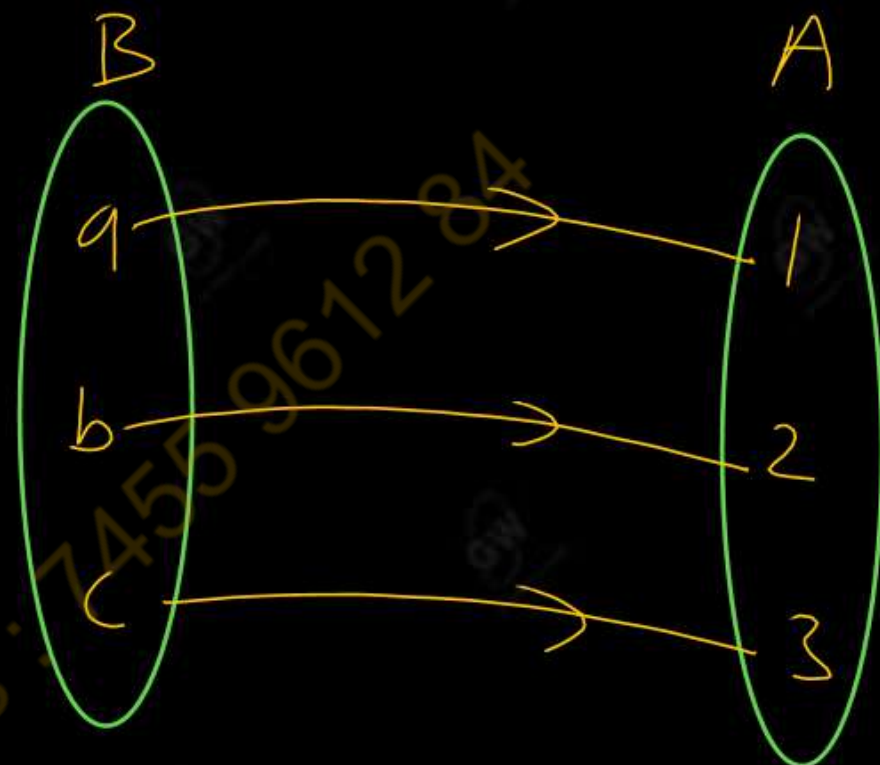
It is not
a function

$$f: A \rightarrow B$$



one-one onto

$$f^{-1}: B \rightarrow A$$



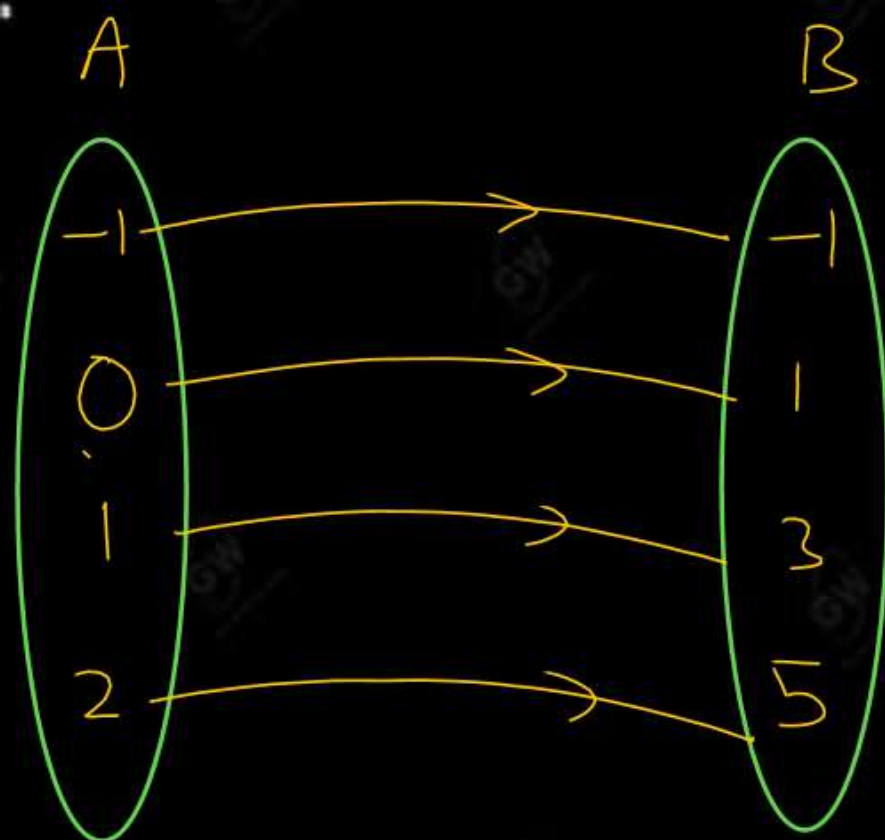
one-one onto

Conclusion : (i) A function $f: A \rightarrow B$ is invertible if it is one one onto function

(ii) If $f: A \rightarrow B$ is ONE-ONE ONTO, then $f^{-1}: B \rightarrow A$ is also ONE-ONE ONTO

(iii) If $f: A \rightarrow B$ is ONE-ONE ONTO function, then $f^{-1}: B \rightarrow A$ is a unique function

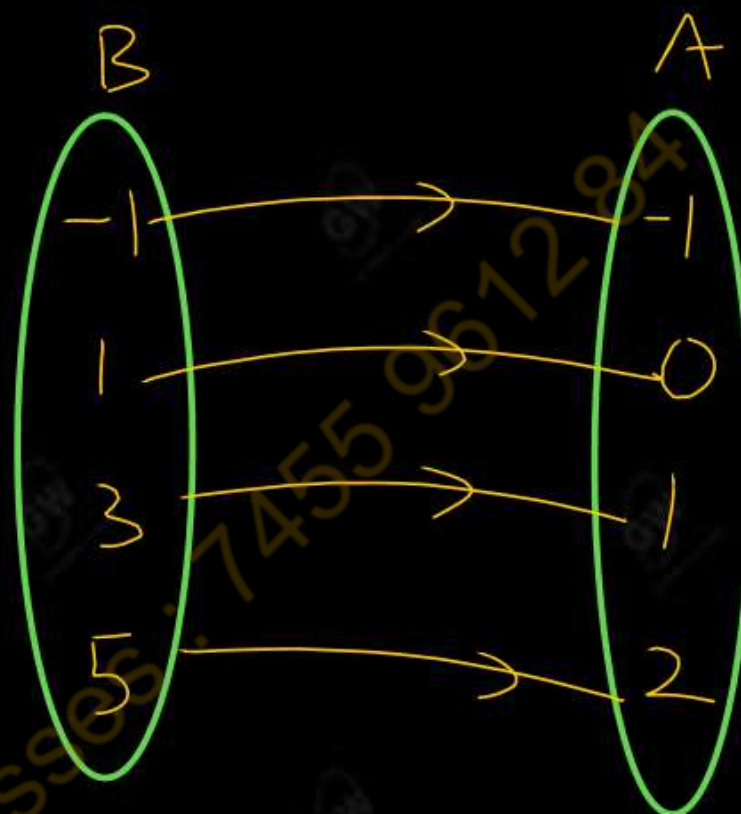
Example: $f(x) = 2x + 1$



$$y = 2x + 1$$

$$2x = y - 1$$

$$x = \frac{y - 1}{2}$$



Steps to find INVERSE

- (i) Put $f(x) = y$, where $y \in B$ and $x \in A$.
- (ii) Solve $f(x) = y$ to obtain x in terms of y .
- (iii) Replace x by $f^{-1}(y)$ to obtain the required inverse of f

$$f^{-1}(y) = \frac{y - 1}{2}$$

Q.1:- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 3x - 7$. Show that f is invertible and hence find the inverse of f .

$$f(x) = 3x - 7$$

$$\text{Let } x_1, x_2 \in \mathbb{R}$$

$$\text{Put } f(x_1) = f(x_2)$$

$$3x_1 - 7 = 3x_2 - 7$$

$$3x_1 = 3x_2$$

$$x_1 = x_2$$

Hence

$f(x)$ is one-one function

Again

$$f(x) = 3x - 7$$

$$\text{Put } f(x) = y$$

$$y = 3x - 7$$

$$3x = y + 7$$

$$x = \frac{y + 7}{3} \quad \text{--- (1)}$$

$$\Rightarrow y \in \mathbb{R}$$

For all $y \in \mathbb{R}, x \in \mathbb{R}$

$$\Rightarrow \text{Range} = \text{codomain}$$

Hence $f(x)$ is an onto function

$\Rightarrow f(x)$ is one-one onto

Replace x by $f^{-1}(y)$ in (1)

$$f^{-1}(y) = \frac{y + 7}{3}$$

Q.2:- Show that $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R} - \{0\}$ given by $f(x) = \frac{1}{x}$, is invertible and find f^{-1} .

$$f(x) = \frac{1}{x}$$

$$\text{Let } x_1, x_2 \in \mathbb{R} - \{0\}$$

$$\text{Put } f(x_1) = f(x_2)$$

$$\frac{1}{x_1} = \frac{1}{x_2}$$

$$x_1 = x_2$$

$\Rightarrow f(x)$ is one-one

Again

$$f(x) = \frac{1}{x}$$

$$y = \frac{1}{x}$$

$$x = \frac{1}{y} \quad \text{--- (1)}$$

$$y \in \mathbb{R} - \{0\}$$

For all $y \in \mathbb{R} - \{0\}$, $x \in \mathbb{R} - \{0\}$

Range $f = \text{codomain } f$

$\Rightarrow f(x)$ is an onto function

$\Rightarrow f(x)$ is one-one onto

Replace x by $f^{-1}(y)$ in (1)

$$f^{-1}(y) = \frac{1}{y}$$

Q.3:- Is the function $f : N \rightarrow N$ defined by $f(x) = x^2$, $x \in N$ is invertible

$$f(n) = n^2$$

$$\text{Let } n_1, n_2 \in N$$

$$\text{Put } f(n_1) = f(n_2)$$

$$n_1^2 = n_2^2$$

$$\Rightarrow n_1 = n_2$$

$\Rightarrow f(n)$ is one-one

$$f(n) = n^2$$

$$\text{Put } f(n) = y$$

$$y = n^2$$

$$n^2 = y$$

$$n = \sqrt{y}$$

For all $y \in N$

$$n \notin N$$

$\Rightarrow \text{Range } f = \text{codomain } f$

$\Rightarrow f(n)$ is not onto

Hence

$f(n)$ is not invertible

Q.4:- Let a function is defined as $f: R - \{3\} \rightarrow R - \{1\}$, $f(x) = (x-2)/(x-3)$, then show that f is a bijective function and also compute the inverse of f . Where R is a set of real numbers.

$$f(x) = \frac{x-2}{x-3}$$

Let $x_1, x_2 \in R - \{3\}$

$$\text{Put } f(x_1) = f(x_2)$$

$$\frac{x_1-2}{x_1-3} = \frac{x_2-2}{x_2-3}$$

$$(x_1-2)(x_2-3) = (x_2-2)(x_1-3)$$

$$\cancel{x_1 x_2} - 3x_1 - 2x_2 + 6 = \cancel{x_1 x_2} - 3x_2 - 2x_1 + 6$$

$$-3x_1 - 2x_2 = -3x_2 - 2x_1$$

$$-3x_1 + 2x_1 = -3x_2 + 2x_2$$

$$-x_1 = -x_2$$

$$x_1 = x_2$$

$\Rightarrow f(x)$ is one-one

Again

$$f(x) = \frac{x-2}{x-3}$$

$$y = \frac{x-2}{x-3}$$

$$xy - 3y = x - 2$$

$$xy - x = 3y - 2$$

$$x(y-1) = 3y-2$$

$$x = \frac{3y-2}{y-1} \quad \text{--- (1)}$$

$$y \in R - \{1\}$$

For all $y \in \mathbb{R} - \{1\}$

$x \in \mathbb{R} - \{3\}$

Range of function = codomain of function

$\Rightarrow f(x)$ is onto

$\Rightarrow f(x)$ is one-one onto function

Replace x by $f^{-1}(y)$ in ①

$$f^{-1}(y) = \frac{3y-2}{y-1}$$

Q.5:- Find the inverse of function $f(x) = \frac{2x}{x-1}$

$$f(n) = \frac{2n}{n-1}$$

Put $f(n) = y$

$$y = \frac{2n}{n-1}$$

$$y(n-1) = 2n$$

$$ny - y = 2n$$

$$ny - 2n = y$$

$$n(y-2) = y$$

$$n = \frac{y}{y-2}$$

Replace n by $f^{-1}(y)$

$$f^{-1}(y) = \frac{y}{y-2}$$

$$f: \mathbb{R} - \{1\} \rightarrow \mathbb{R} - \{2\}$$

Topic : Inverse of a function

Q.1. Let $f : R \rightarrow R$ is defined by $f(x) = ax + b$ where $a, b \in R$ and $a \neq 0$. Show that f is invertible and find the inverse of f .

Q.2. Show that $f : R - \{0\} \rightarrow R - \{0\}$, given by $f(x) = \frac{3}{x}$ is invertible and find f^{-1} .

Q.3. Let $A = R - \{3\}$ and $B = R - \{1\}$ Consider the function $f: A \rightarrow B$, defined by

$$f(x) = \frac{x-2}{x-3} \text{ Show that } f \text{ is invertible and find } f^{-1}.$$

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Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS

(Discrete Mathematics)

UNIT - 2 : Functions and Boolean Algebra

Lecture - 03

Today's Target

- Composition of Functions
- PYQ
- DPP

Composition of Functions

Let A , B and C be three non-empty set and let

$$f : A \rightarrow B \text{ and } g : B \rightarrow C$$

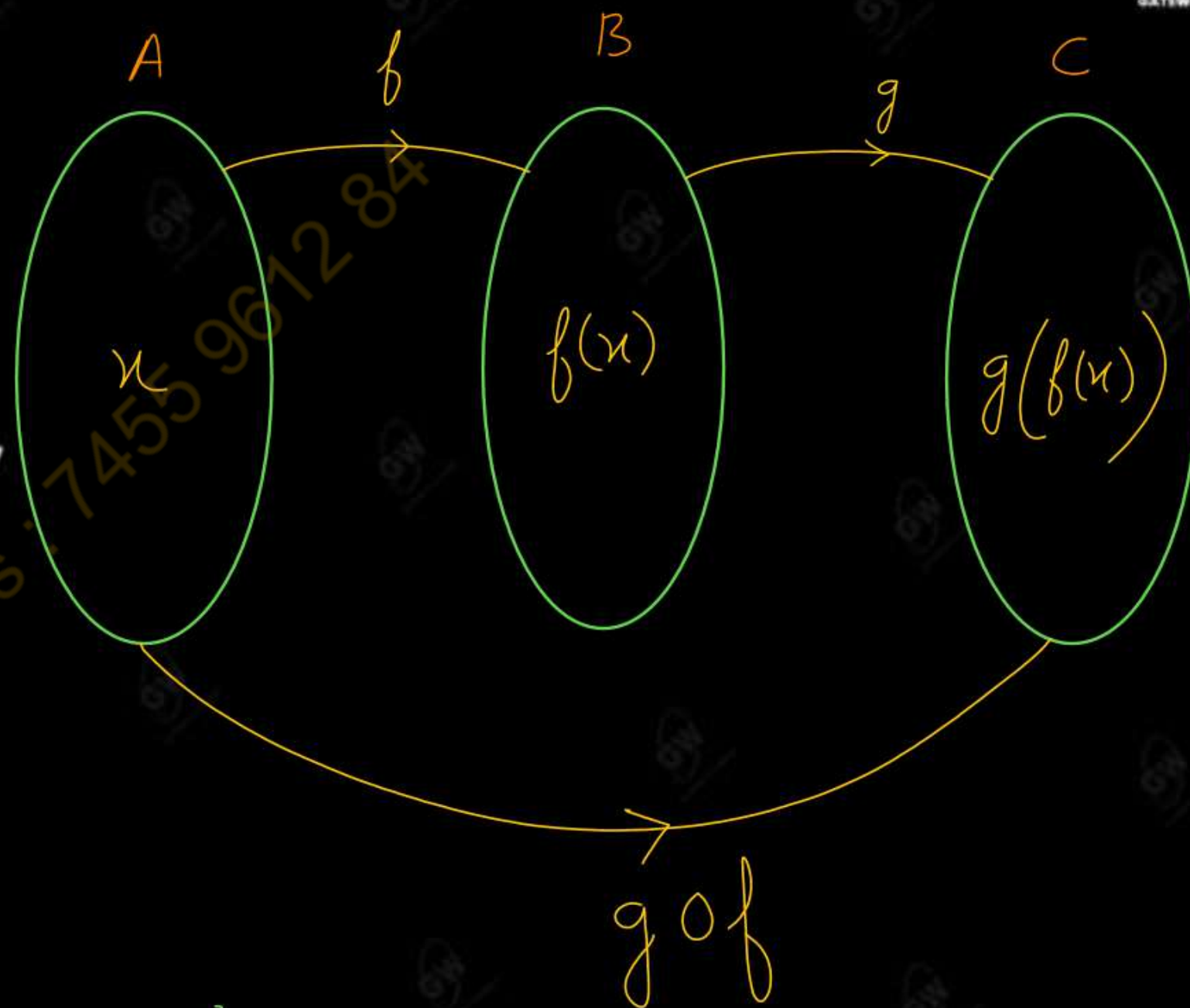
be two functions

Then the composition of f and g is a function denoted by

$$g \circ f : A \rightarrow C$$

And is defined as

$$g \circ f (x) = g\{f(x)\} \quad \forall x \in A$$



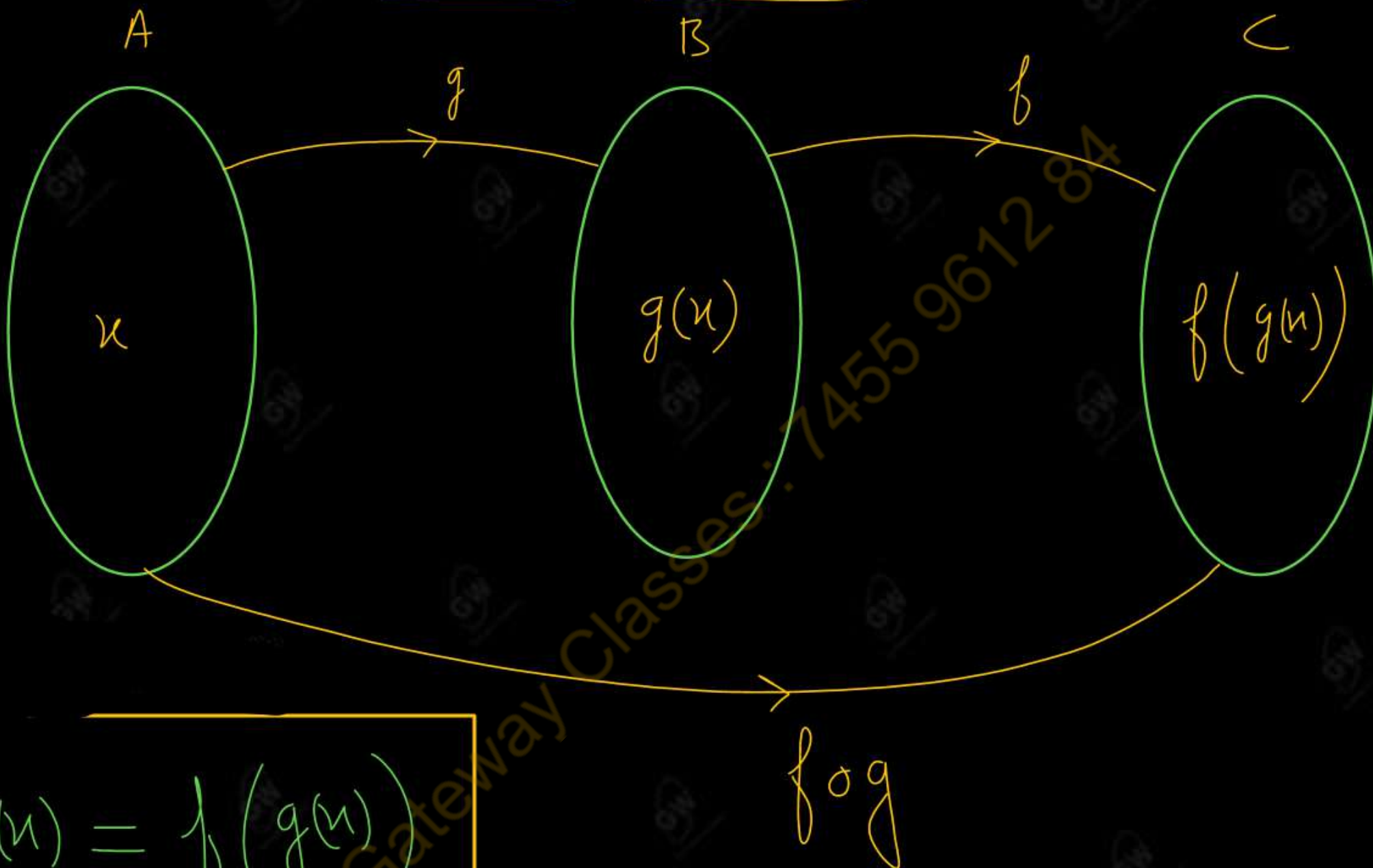
Note:- $g \circ f$ exist if and only if Range of $f \subseteq$ domain of g

Note:- $f \circ g$ exists if and only if, $\text{Range of } g \subseteq \text{domain of } f$

$$g: A \rightarrow B$$

$$f: B \rightarrow C$$

$$f \circ g: A \rightarrow C$$



$$(f \circ g)(u) = f(g(u))$$

Q.1:- Let $f = \overset{A}{\{1, 3, 4\}} \rightarrow \overset{B}{\{1, 2, 5\}}$ and $g = \overset{B}{\{1, 2, 5\}} \rightarrow \overset{C}{\{1, 3\}}$ be defined as $f = \{(1, 2)(3, 5)(4, 1)\}$ and $g = \{(1, 3)(2, 3)(5, 1)\}$, Find $g \circ f$ and $f \circ g$.

Given

$$f = \{1, 3, 4\} \rightarrow \{1, 2, 5\}$$

$$g = \{1, 2, 5\} \rightarrow \{1, 3\}$$

$$f = \{(1, 2)(3, 5)(4, 1)\}$$

$$g = \{(1, 3)(2, 3)(5, 1)\}$$

$$(i) \text{Rang}(f) = \{1, 2, 5\}$$

$$\text{Dom}(g) = \{1, 2, 5\}$$

$$\Rightarrow \text{Rang}(f) \subseteq \text{Dom}(g)$$

$$\Rightarrow g \circ f \text{ exists}$$

$$\text{dom}(g \circ f) = \text{dom}(f)$$

$$\text{dom}(g \circ f) = \{1, 3, 4\}$$

$$(g \circ f)(1) = g(f(1))$$

$$= g(2) = 3$$

$$(g \circ f)(3) = g(f(3))$$

$$= g(5)$$

$$= 1$$

$$(g \circ f)(4) = g(f(4))$$

$$= g(1)$$

$$= 3$$

$$g \circ f = \{(1, 3), (3, 1), (4, 3)\}$$

$$(ii) \text{ rang}(g) = \{1, 3\}$$

$$\text{dom}(f) = \{1, 3, 4\}$$

$$\Rightarrow \text{rang}(g) \subseteq \text{dom}(f)$$

$$\Rightarrow f \circ g \text{ exists}$$

$$\text{dom}(f \circ g) = \text{dom}(g)$$

$$\text{dom}(f \circ g) = \{1, 2, 5\}$$

$$(f \circ g)(1) = f(g(1)) = f(3) = 5$$

$$(f \circ g)(2) = f(g(2)) = f(3) = 5$$

$$(f \circ g)(5) = f(g(5)) = f(1) = 2$$

$$f \circ g = \{(1, 5), (2, 5), (5, 2)\}$$

Q.2:- Find the composition mapping $g \circ f$ if $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = e^x$ and

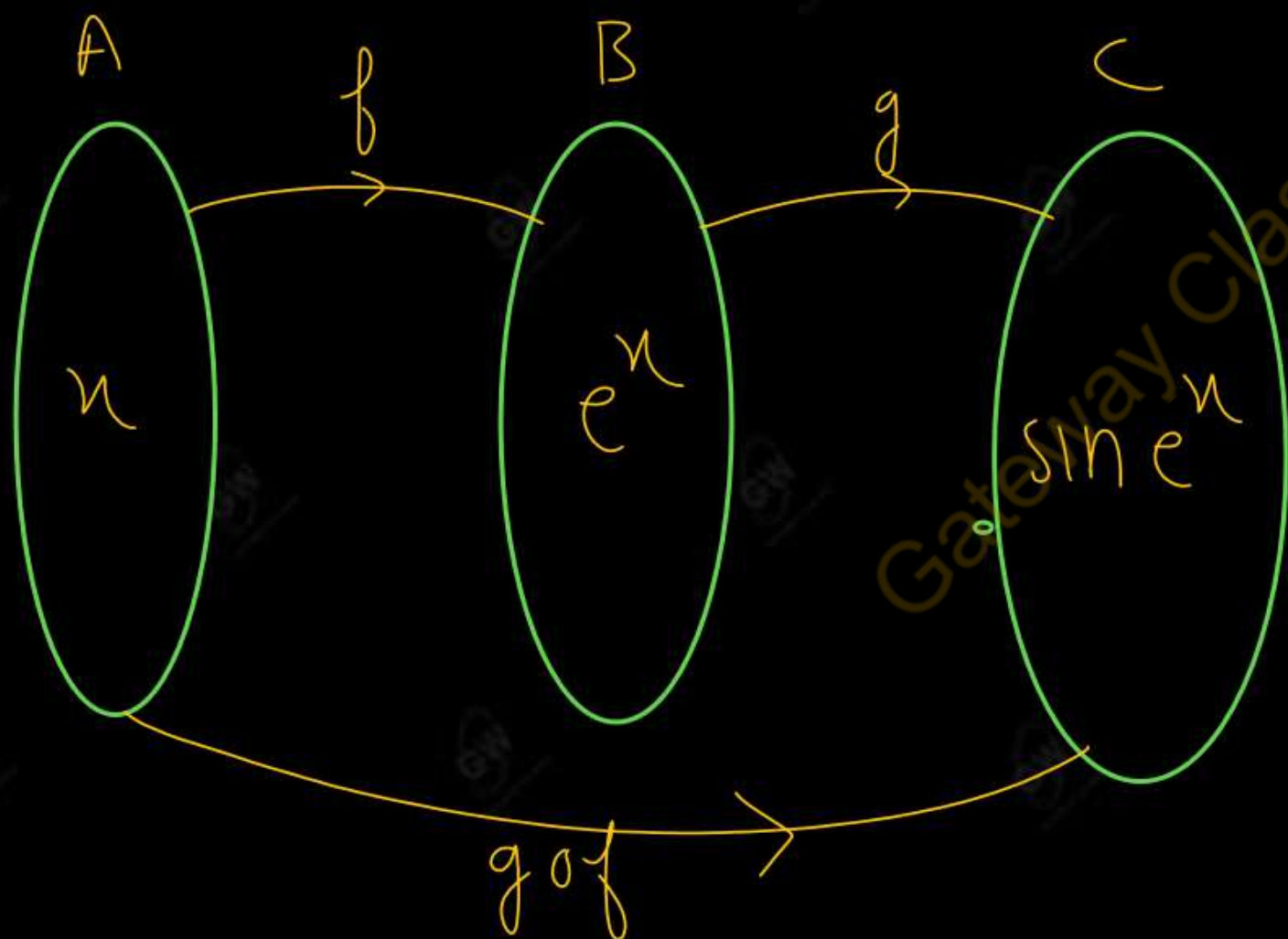
(AKTU-2022-23)

$g : \mathbb{R} \rightarrow \mathbb{R}$ is given by $g(x) = \sin x$.

Given

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$$

$$g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = \sin x$$



$$g \circ f(x) = g(f(x))$$

$$= g(e^x)$$

$$g \circ f(x) = \sin e^x$$

Q.3:- Let $A = \{1, 2, 3\}$

$B = \{a, b\}$

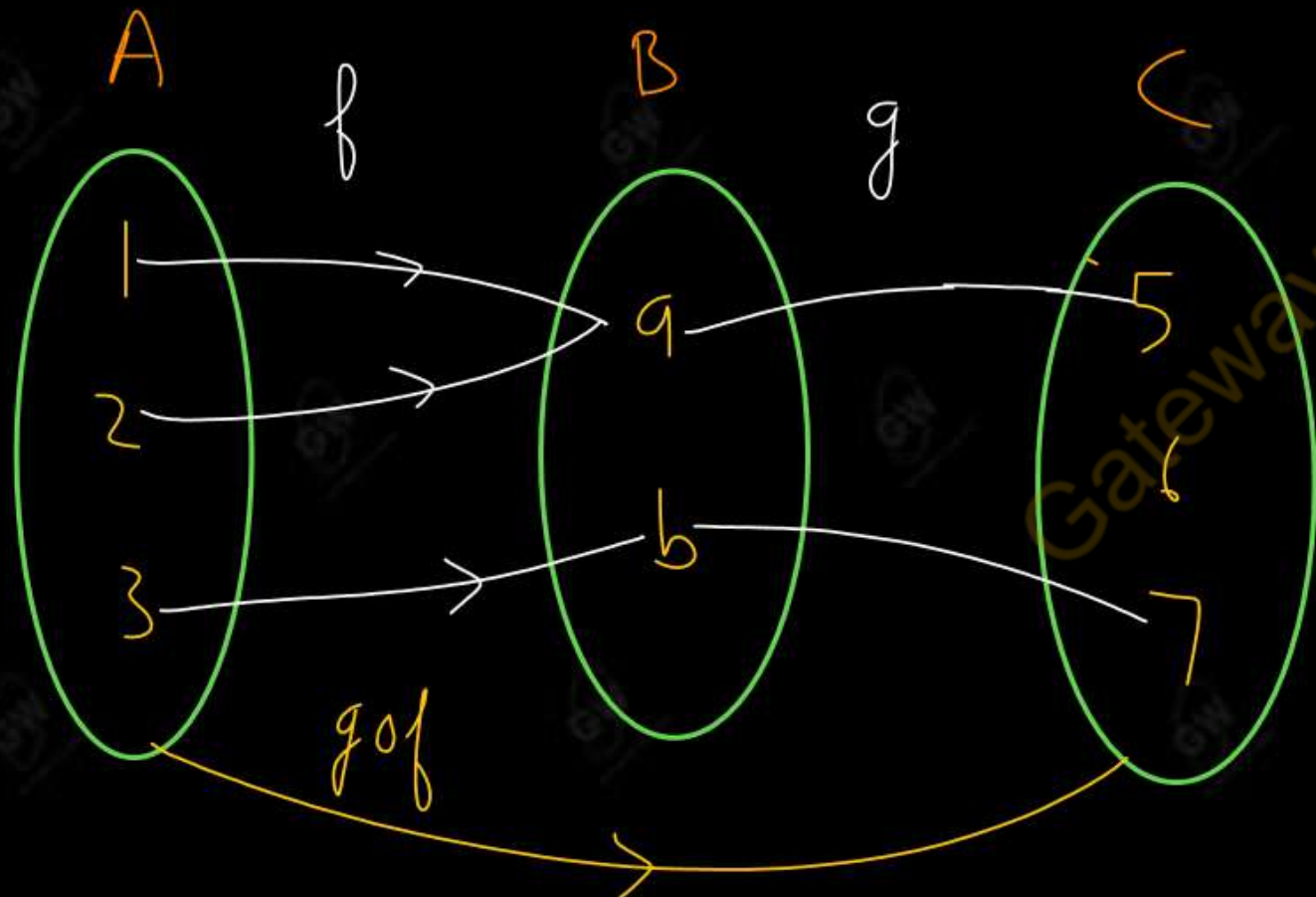
$C = \{5, 6, 7\}$

Define $f : A \rightarrow B$ where $f = \{(1, a) (2, a) (3, b)\}$

and $g : B \rightarrow C$ where $g = \{(a, 5) (b, 7)\}$

Find the composition of f and g .

1st Method



$$g \circ f = \{(1, 5) (2, 5) (3, 7)\}$$

2nd Method

$$\begin{aligned} \text{dom}(g \circ f) &= \text{dom}(f) \\ &= \{1, 2, 3\} \end{aligned}$$

$$(g \circ f)(1) = g(f(1)) = g(a) = 5$$

$$(g \circ f)(2) = g(f(2)) = g(a) = 5$$

$$(g \circ f)(3) = g(f(3)) = g(b) = 7$$

$$g \circ f = \{(1, 5) (2, 5) (3, 7)\}$$

Q.4:- Let $f : R \rightarrow R$ and $g : R \rightarrow R$ where R be the Set of real numbers.

Find (i) $g \circ f$ (ii) $f \circ g$ (iii) $f \circ f$ (iv) $g \circ g$. Where $f(x) = x^2$ and $g(x) = x + 4$

Given

$$f : R \rightarrow R, \quad f(x) = x^2$$

$$g : R \rightarrow R, \quad g(x) = x + 4$$

$$(i) \quad g \circ f = (g \circ f)(x)$$

$$= g(f(x))$$

$$= g(x^2)$$

$$= x^2 + 4$$

$$(ii) \quad f \circ g = (f \circ g)(x)$$

$$= f(g(x))$$

$$= f(x + 4)$$

$$= (x + 4)^2$$

$$(iii) \quad f \circ f = (f \circ f)(x)$$

$$= f(f(x))$$

$$= f(x^2)$$

$$f \circ f = (x^2)^2 = x^4$$

$$(iv) \quad g \circ g = g \circ g(x)$$

$$= g(g(x))$$

$$= g(x + 4)$$

$$= x + 4 + 4$$

$$= x + 8$$

Q.5:- If $f : R \rightarrow R$ and $g : R \rightarrow R$ $h : R \rightarrow R$ defined by

$$f(x) = 3x^2 + 2, \quad g(x) = 7x - 5 \quad \text{and} \quad h(x) = \frac{1}{x}.$$

Find the following composition functions.

(a) $(f \circ g \circ h)(x)$

(b) $(h \circ g \circ f)(x)$

(c) $(g \circ g)(x)$

(d) $(g \circ h)(x)$

Given

$$f : R \rightarrow R, \quad f(x) = 3x^2 + 2$$

$$g : R \rightarrow R, \quad g(x) = 7x - 5$$

$$h : R \rightarrow R, \quad h(x) = \frac{1}{x}$$

$$(i) (f \circ g \circ h)(x) = f\left(g\left(h(x)\right)\right)$$

$$= f\left(g\left(\frac{1}{x}\right)\right) = f\left(7 \times \frac{1}{x} - 5\right)$$

$$= 3\left(\frac{7}{x} - 5\right)^2 + 2$$

$$(ii) (h \circ g \circ f)(x) = h(g(f(x)))$$

$$= h(g(3x^2 + 2))$$

$$= h(7(3x^2 + 2) - 5)$$

$$= h(21x^2 + 14 - 5)$$

$$= h(21x^2 + 9)$$

$$= \frac{1}{21x^2 + 9}$$

$$(iii) g \circ g(x) = g(g(x)) = g(7x - 5)$$

$$= 7(7x - 5) - 5$$

$$= 49x - 40$$

$$(iv) g \circ h(x) = g(h(x)) = g\left(\frac{1}{x}\right)$$

$$= \frac{7}{x} - 5$$

Q.6:- Let $X = \{a, b, c\}$. Define $f : X \rightarrow X$ such that

$$f = \{(a, b), (b, a), (c, c)\}$$

Find

(i) f^2

(ii) f^3

(iii) f^4

Given

$$X = \{a, b, c\}$$

$$f = \{(a, b), (b, a), (c, c)\}$$

$$\begin{aligned} \text{(i)} \quad f^2(a) &= (f \circ f)(a) \\ &= f(f(a)) \\ &= f(b) = a \end{aligned}$$

$$\begin{aligned} f^2(b) &= (f \circ f)(b) \\ &= f(f(b)) \\ &= f(a) \\ &= b \end{aligned}$$

$$\begin{aligned} f^2(c) &= (f \circ f)(c) \\ &= f(f(c)) \\ &= f(c) \\ &= c \end{aligned}$$

$$f^2 = f \circ f$$

(UPTU-2014)

$$f^2(c) = c$$

$$f^2 = \{(a, a), (b, b), (c, c)\}$$

$$\begin{aligned} \text{(ii)} \quad f^3(a) &= (f^2 \circ f)(a) \\ &= f^2(f(a)) \\ &= f^2(b) \\ &= a \end{aligned}$$

$$f^3(b) = (f^2 \circ f)(b)$$

$$= f^2(f(b))$$

$$= f^2(a)$$

$$= a$$

$$f^3(c) = (f^2 \circ f)(c)$$

$$= f^2(f(c))$$

$$= f^2(c)$$

$$= c$$

$$f^3 = \{(a, a), (b, a), (c, c)\}$$

$$(iii) f^4(a) = (f^3 \circ f)(a)$$

$$= f^3(f(a))$$

$$= f^3(a)$$

$$= a$$

$$f^4(b) = (f^3 \circ f)(b)$$

$$= f^3(f(b))$$

$$= f^3(a)$$

$$= a$$

$$f^4(c) = (f^3 \circ f)(c)$$

$$= f^3(f(c))$$

$$= f^3(c)$$

$$= c$$

$$f^4 = \{(a, a), (b, a), (c, c)\}$$

Theorem:- if $f : A \rightarrow B$ and $g : B \rightarrow C$ be one - one and onto function. Proof that
 $g \circ f : A \rightarrow C$ is one - one onto

And

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Given

$f : A \rightarrow B$ is one-one onto

$g : B \rightarrow C$ is also one-one onto

Proof: since f and g are one
one onto function

$\therefore f^{-1}$ and g^{-1} both are exist

$$\text{If } y = f(x) \Rightarrow f^{-1}(y) = x \quad \text{--- (1)}$$

$$\text{If } z = g(y) \Rightarrow g^{-1}(z) = y \quad \text{--- (2)}$$

For one-one

Let $x_1, x_2 \in A$

$$(g \circ f)(x_1) = (g \circ f)(x_2)$$

$$g(f(x_1)) = g(f(x_2))$$

$$g(f(n_1)) = g(f(n_2))$$

since g is one-one

$$f(n_1) = f(n_2)$$

since f is one-one

$$n_1 = n_2$$

$\Rightarrow g \circ f$ is one-one function

For onto

Let $z \in C$

— since g is an onto function, then

there exist an element $y \in B$ such

$$\text{that } g(y) = z$$

— since f is an onto function, then

there exist an element $x \in A$, such

$$\text{that } f(x) = y$$

$$\Rightarrow z = g(y) = g(f(u))$$

$$z = g \circ f(u)$$

Hence, $g \circ f$ is an onto function

$\Rightarrow g \circ f$ is an one-one onto function

$\Rightarrow (g \circ f)^{-1}$ exist

$$(g \circ f)^{-1}(z) = u \quad \text{--- (3)}$$

Now

$$(f^{-1} \circ g^{-1})(z) = f^{-1}(g^{-1}(z))$$

$$= f^{-1}(y) \quad \{ \text{From (2)} \}$$

$$= u \quad \{ \text{From (1)} \}$$

$$(f^{-1} \circ g^{-1})(z) = u \quad \text{--- (4)}$$

From (3) and (4)

$$(g \circ f)^{-1}(z) = (f^{-1} \circ g^{-1})(z)$$

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Topic : Composition of Functions

Q.1 Let $X = \{1, 2, 3\}$, $Y = \{q, p\}$ and $Z = \{a, b\}$, let $f: X \rightarrow Y$ be $f = \{(1, p), (2, p), (3, q)\}$

$g: Y \rightarrow Z$ be $g = \{(p, b), (q, b)\}$. Find $g \circ f$ and show it pictorially.

(UPTU-2013)

$$g \circ f = \{(1, b), (2, b), (3, b)\}$$

Q.2 If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = x^3 - 4x, g(x) = 1 / (x^2 + 1) \quad h(x) = x^4.$$

Find the following composition functions.

(a) $(f \circ g \circ h)(x)$

(b) $(h \circ g \circ f)(x)$

(c) $(g \circ g)(x)$

(d) $(g \circ h)(x)$

$$(c) (x^2 + 1)^2 [1 + (x^2 + 1)^2] \quad (d) (x^8 + 1)^{-1}$$

Ans (a) $(x^8 + 1)^{-3} - 4(x^8 + 1)^{-1}$

(b) $[(x^3 - 4x)^2 + 1]^{-4}$

Q.3 If $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $g : \mathbb{Z} \rightarrow \mathbb{Z}$, $h : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(n) = n^2$, $g(n) = n + 1$ and $h(n) = n - 1$. Find

(a) $g \circ f \circ h$

(b) $f \circ g \circ h$

(c) $h \circ f \circ g$.

Ans (a) $n^2 - 2n + 2$ (b) n^2 (c) $n^2 + 2n$

Q.4 Given $f = \{(a, b), (b, a), (c, b)\}$ a function from $X = \{a, b, c\}$ to X :

(a) Write $f \circ f$ and $f \circ f \circ f$ as sets of ordered pairs. Ans $f \circ f = \{(a, a), (b, b), (c, a)\}$

(b) Define $f^n = f \circ f \circ \dots \dots \dots$ of to be the n - fold composition of f with itself. Find f^9 .

$$f \circ f \circ f = \{(a, b), (b, a), (c, d)\}$$

$$(b) \quad f^9 = f$$

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Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS

(Discrete Mathematics)

UNIT -2 : Functions and Boolean Algebra

Lecture - 04

Today's Target

- **Definitions of Boolean Algebra**
- **DPP**

Boolean Algebra is an algebraic structure which is based on the principle of logics.

Algebraic Structure for Boolean Algebra

✓ (i) $(B, +, \cdot, ', 0, 1)$

(ii) $(B, \vee, \wedge, ', 0, 1)$

(iii) $(B, \cup, \cap, ', 0, 1)$

First Definition:

A non-empty set B with two binary operations $+$ and $\cdot$, a unary operation $'$, and two distinct elements 0 and 1 is called a Boolean algebra, denoted by $(B, +, \cdot, ', 0, 1)$ if and only if the following properties are satisfied

1. Closure law : $\forall a, b \in B$

$$(i) a + b \in B$$

$$(ii) a \cdot b \in B$$

2. Commutative law : $\forall a, b \in B$

$$(i) a + b = b + a$$

$$(ii) a \cdot b = b \cdot a$$

3. Distributive law : $\forall a, b, c \in B$

$$(i) a \cdot (b + c) = a \cdot b + a \cdot c$$

$$(ii) a + (b \cdot c) = (a + b) \cdot (a + c)$$

$$(b + c) \cdot a = b \cdot a + c \cdot a$$

$$(b \cdot c) + a = (b + a) \cdot (c + a)$$

✓ **4. Identity law :** $\forall a \in B$

(i) $a + 0 = 0 + a = a$



Additive Identity

(ii) $a \cdot 1 = 1 \cdot a = a$



Multiplicative Identity

✓✓ **5. Complement law :** $\forall a \in B$ there exist $a' \in B$ such that

(i) $a + a' = a' + a = 1$ (ii) $a \cdot a' = a' \cdot a = 0$

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Q1. Show that the structure $(B, +, \cdot, ')$ is a Boolean algebra where $B = \{0, 1\}$ and $+$ and $\cdot$ are two binary operation and $'$ a unary operation on B , defined by the following table

$+$	0	1
0	0	1
1	1	1

(i)

$\cdot$	0	1
0	0	0
1	0	1

(ii)

a	a'
0	1
1	0

(iii)

Sol

① closure law : $B = \{0, 1\}$

(i) $0 + 1 = 1 \in B$ Hence closure satisfied under $+$ and $\cdot$.

(ii) $0 \cdot 1 = 0 \in B$

② Commutative law: $\forall 0, 1 \in B$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$\Rightarrow 0 + 1 = 1 + 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$0 \cdot 1 = 1 \cdot 0$$

Hence commutative is satisfied under '+' and '•'

③ Distributive law: $\forall a, b, c \in B$

$$(i) a \cdot (b + c) = a \cdot b + a \cdot c$$

$$(ii) a + (b \cdot c) = (a + b) \cdot (a + c)$$

$$\text{Let } a = 1, b = 0, c = 1$$

$$(i) a \cdot (b + c) = 1 \cdot (0 + 1)$$

$$= 1 \cdot 1$$

$$= 1$$

$$a \cdot b + a \cdot c = 1 \cdot 0 + 1 \cdot 1$$

$$= 0 + 1$$

$$= 1$$

$$\Rightarrow a \cdot (b + c) = a \cdot b + a \cdot c$$

$$(ii) a + (b \cdot c) = 1 + (0 \cdot 1)$$

$$= 1 + 0$$

$$= 1$$

$$\begin{aligned}
 (a+b).(a+c) &= (1+0).(1+1) \\
 &= 1.1 \\
 &= 1
 \end{aligned}$$

$$\Rightarrow a + (b.c) = (a+b).(a+c)$$

Hence Distributive law is satisfied

④ Identity law

$$0 + 0 = 0$$

$$1 + 0 = 1$$

$$0.1 = 0$$

$$1.1 = 1$$

Hence, 0 is additive identity

1 is multiplicative identity

⑤ Complement law: For $0, 1 \in B$

$$1' = 0 \in B$$

$$0' = 1 \in B$$

$\Rightarrow$ complement law is satisfied

Hence $(B, +, \cdot, ', 0, 1)$ is a Boolean Algebra

Second Definition:

A complemented distributive lattice is called a Boolean algebra.

OR

A non-empty set B with two binary operations $\vee$ and $\wedge$, a unary operation $'$, and two distinct elements 0 and 1 is called a Boolean algebra, denoted by $(B, \vee, \wedge, ', 0, 1)$ if and only if the following properties are satisfied

1. Closure law : $\forall a, b \in B$

$$(i) a \vee b \in B$$

$$(ii) a \wedge b \in B$$

2. Commutative law : $\forall a, b \in B$

$$(i) \underline{a \vee b} = \underline{b \vee a}$$

$$(ii) a \wedge b = b \wedge a$$

3. Distributive law : $\forall a, b, c \in B$

$$(i) a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$(ii) a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

4. Identity law : There exist two elements 0 and 1 in B such that

$$(i) a \vee 0 = a$$

$$(ii) a \wedge 1 = a$$

$$\forall$$

5. Complement law : $\forall a \in B$ there exist an element $a' \in B$ such that

$$(i) a \vee a' = 1$$

$$(ii) a \wedge a' = 0$$

Q. 2: Let B be the set of all +ve divisors of 30 and the operations $\vee$ and $\wedge$ on B are defined as follows

$$a \vee b = \text{LCM}(a, b)$$

$$a \wedge b = \text{HCF}(a, b)$$

Prove that $(B, \vee, \wedge, ')$ is a Boolean algebra

Solution

$$B = \{1, 2, 3, 5, 6, 10, 15, 30\}$$

↑
least element

↑
greatest element

composition table for $\vee$

$\vee$	1	2	3	5	6	10	15	30
1	1	2	3	5	6	10	15	30
2	2	2	6	10	6	10	30	30
3	3	6	3	15	6	30	15	30
5	5	10	15	5	30	10	15	30
6	6	6	6	30	6	30	30	30
10	10	10	30	10	30	10	30	30
15	15	30	15	15	30	30	15	30
30	30	30	30	30	30	30	30	30

Composition table for $\wedge$

$\wedge$	1	2	3	5	6	10	15	30
1	1	1	1	1	1	1	1	1
2	1	2	1	1	2	2	1	2
3	1	1	3	1	3	1	3	3
5	1	1	1	5	1	5	5	5
6	1	2	3	1	6	2	3	6
10	1	2	1	5	2	10	5	10
15	1	1	3	5	3	5	15	15
30	1	2	3	5	6	10	15	30

Composition table for complement

a	a'	
1	30	least element
2	15	$= 1$
3	10	greatest element
5	6	$= 30$
6	5	$a \vee a' = 30$
10	3	$a \wedge a' = 1$
15	2	
30	1	

① Closure law:

Since all the elements of V and $\wedge$ table lie inside the set B

Hence closure law is satisfied under V and $\wedge$

$$\Rightarrow \forall a, b \in B$$

$$a \vee b \in B \text{ and } a \wedge b \in B$$

② Commutative law

Since corresponding row and column are identical in V and $\wedge$ table

$\Rightarrow$ Commutative law is satisfied under V and $\wedge$

$$\Rightarrow \forall a, b \in B$$

$$(i) a \vee b = b \vee a$$

$$(ii) a \wedge b = b \wedge a$$

(3) Distributive law

$$(i) a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$(ii) a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$\text{Let } a = 2, b = 3, c = 6$$

$$(i) a \vee (b \wedge c) = 2 \vee (3 \wedge 6)$$

$$= 2 \vee 3$$

$$= 6$$

$$(a \vee b) \wedge (a \vee c) = (2 \vee 3) \wedge (2 \vee 6)$$

$$= 6 \wedge 6$$

$$= 6$$

$$\Rightarrow a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

Similarly

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

Hence Distributive law is satisfied

under $\vee$ and $\wedge$

(4) Identity law: $\forall a \in B$

From $\vee$ table

$$a \vee 1 = a$$

From $\wedge$ table

$$a \wedge 0 = a$$

Hence

1 is the identity element for $\vee$

0 is

"

"

$\wedge$

(5) Complement law

$\forall a \in B$ there exist

an element $a' \in B$

such that

$$a \vee a' = 1$$

$$a \wedge a' = 0$$

Hence $(B, \vee, \wedge, ', 0, 1)$

Third Definition

A non-empty set B with two binary operations $\cup$ and $\cap$, a unary operation $'$, and two distinct elements 0 and 1 is called a Boolean algebra, denoted by $(B, \cup, \cap, ', 0, 1)$ if and only if the following properties are satisfied

1. Closure law : $\forall A_1, A_2 \in B$

$$(i) A_1 \cup A_2 \in B$$

$$(ii) A_1 \cap A_2 \in B$$

2. Commutative law : $\forall A_1, A_2 \in B$

$$(i) A_1 \cup A_2 = A_2 \cup A_1$$

$$(ii) A_1 \cap A_2 = A_2 \cap A_1$$

3. Distributive law : $\forall A_1, A_2, A_3 \in B$

$$(i) A_1 \cup (A_2 \cap A_3) = (A_1 \cup A_2) \cap (A_1 \cup A_3)$$

$$(ii) A_1 \cap (A_2 \cup A_3) = (A_1 \cap A_2) \cup (A_1 \cap A_3)$$

✓ **4. Identity law :** $\forall A \in B$ there exist two element ϕ and $\cup$ in B such that

$$(i) A \cup \phi = A$$

$$(ii) A \cap \cup = A$$

✓ **5. Complement law :** $\forall A \in B$ there exist an element $A' \in B$ such that

$$(i) A \cup A' = \cup \quad (ii) A \cap A' = \phi$$

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DISCRETE STRUCTURES & THEORY OF LOGICS

(Discrete Mathematics)

UNIT -2 : Functions and Boolean Algebra

Lecture - 05

Today's Target

- Theorems of Boolean Algebra
- PYQ
- DPP

$(B, +, \cdot, ', 0, 1)$ is a Boolean algebra if and only if the following properties are satisfied

1. Closure Laws: $\forall a, b \in B$

(i) $a + b \in B$

(ii) $a \cdot b \in B$

2. Commutative Laws: $\forall a, b \in B$

(i) $a + b = b + a$

(ii) $a \cdot b = b \cdot a$

3. Associative Laws: $\forall a, b, c \in B$

(i) $a + (b + c) = (a + b) + c$

(ii) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$

4. Distributive Laws: $\forall a, b, c \in B$

(i) $a \cdot (b + c) = a \cdot b + a \cdot c$

(ii) $a + (b \cdot c) = (a + b) \cdot (a + c)$

$(b + c) \cdot a = b \cdot a + c \cdot a$

$(b \cdot c) + a = (b + a) \cdot (c + a)$

5. Identity Law: $\forall a \in B$

(i) $a + 0 = 0 + a = a$

Additive Identity

(ii) $a \cdot 1 = 1 \cdot a = a$

Multiplicative Identity

6. Complement Law: $\forall a \in B$ there exist $a' \in B$ such that

(i) $a + a' = a' + a = 1$

(ii) $a \cdot a' = a' \cdot a = 0$

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Theorems 1 - (Idempotent Law) : For every element of a Boolean algebra B

(i) $a + a = a$

(i) To Prove: $a + a = a$

Proof: RHS

$$a = a + 0 \quad \text{by identity law}$$

$$= a + a \cdot a' \quad \text{by complement law}$$

$$= (a + a) \cdot (a + a') \quad \text{by distributive law}$$

$$= (a + a) \cdot 1 \quad \text{by complement law}$$

$$= a + a \quad \text{by identity law}$$

$$= \text{RHS} \quad \text{Hence proved}$$

(ii) $a \cdot a = a$

(ii) To Prove: $a \cdot a = a$

Proof: LHS

$$a \cdot a = a \cdot a + 0 \quad \text{by identity law}$$

$$= a \cdot a + a \cdot a' \quad \text{by complement law}$$

$$= a \cdot (a + a') \quad \text{by distributive law}$$

$$= a \cdot 1 \quad \text{by complement law}$$

$$= a \quad \text{by identity law}$$

$$= \text{RHS}$$

Hence proved

(i) $a + 1 = 1$

(i) To Prove: $a + 1 = 1$

LHS:

$$\begin{aligned} a + 1 &= a + (a + a') \text{ by complement law} \\ &= (a + a) + a' \text{ by associative law} \\ &= a + a' \text{ by idempotent law} \\ &= 1 \text{ by complement} \end{aligned}$$

$= \text{RHS}$

Hence proved

(ii) $a \cdot 0 = 0$

To Prove : $a \cdot 0 = 0$

Proof : LHS

$$\begin{aligned} a \cdot 0 &= a \cdot (a \cdot a') \text{ by complement law} \\ &= (a \cdot a) \cdot a' \text{ by Associative law} \\ &= a \cdot a' \text{ by idempotent law} \\ &= 0 \text{ by complement law} \end{aligned}$$

$= \text{RHS}$

Hence proved

$$(i) a + a \cdot b = a$$

(i) To Prove: $a + a \cdot b = a$

Proof: LHS

$$a + a \cdot b = a \cdot 1 + a \cdot b \text{ By identity law}$$

$$= a \cdot (1 + b) \text{ By distributive law}$$

$$= a \cdot 1 \text{ By Boundedness law}$$

$$= a \text{ By identity law}$$

$$= \text{RHS}$$

Hence proved

$$(ii) a \cdot (a + b) = a$$

(ii) To Prove: $a \cdot (a + b) = a$

Proof: LHS

$$a \cdot (a + b) = (a + 0) \cdot (a + b) \text{ by identity law}$$

$$= a + (0 \cdot b) \text{ by distributive law}$$

$$= a + 0 \text{ by boundedness law}$$

$$= a \text{ by identity law}$$

$$= \text{RHS}$$

Hence proved

i.e for $0, 1 \in B$ we have

(i) To Prove : $0' = 1$

(i) $0' = 1$

(ii) $1' = 0$

Proof :

$$\begin{aligned} \text{LHS} &= 0' \\ &= 0' + 0 \quad \text{by identity law} \\ &= 1 \quad \text{by complement law} \\ &= \text{RHS} \end{aligned}$$

Hence proved

(ii) To Prove : $1' = 0$

Proof :

$$\begin{aligned} \text{LHS} &= 1' \\ &= 1' \cdot 1 \quad \text{by identity law} \\ &= 0 \quad \text{by complement law} \\ &= \text{RHS} \end{aligned}$$

Hence proved

OR

For each $a \in B$, a' is unique

Proof: Let $a \in B$ be any element

Suppose x and y are two complements of a

By complement law

$$a + x = x + a = 1 \quad \text{and} \quad a \cdot x = x \cdot a = 0 \quad \text{--- (1)}$$

$$a + y = y + a = 1 \quad \text{and} \quad a \cdot y = y \cdot a = 0 \quad \text{--- (2)}$$

NOW

$$x = x \cdot 1$$

By identity law

$$x = x \cdot (a + y)$$

From (2)

$$= x \cdot a + x \cdot y$$

By Distributive law

$$= 0 + x \cdot y$$

From (1)

$$= a \cdot y + x \cdot y$$

From (2)

$$= (a + x) \cdot y$$

By distributive law

$$= 1 \cdot y$$

From (1)

$$\boxed{x = y}$$

By identity law

Hence, complement of each element is unique

OR

The elements 0 and 1 are unique in Boolean algebra.

(i) To Prove identity element 0 is unique for +

Suppose 0 and 0_1 are two identity elements of Boolean algebra B for +

If 0 is the identity element

$$0_1 + 0 = 0_1 \quad \text{--- (1)}$$

If 0_1 is the identity element

$$0 + 0_1 = 0 \quad \text{--- (2)}$$

By commutative law

$$0_1 + 0 = 0 + 0_1$$

$$\Rightarrow 0 = 0_1 \quad (\text{From (1) and (2)})$$

Hence identity element 0 is unique

similarly, we can prove identity element 1 is unique for $\cdot$.

Theorems 7 : Involution Law

For each element a of Boolean algebra B

$$(a')' = a$$

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(i) $(a + b)' = a' \cdot b'$

To Prove : $(a + b)' = a' \cdot b'$

Proof : To show the complement of $(a + b)$ is $a' \cdot b'$, it is sufficient to prove

$(a + b) + a' \cdot b' = 1$ ——— (1)

$(a + b) \cdot (a' \cdot b') = 0$ ——— (2)

LHS of (1)

$(a + b) + a' \cdot b' = (a + b + a') \cdot (a + b + b')$

By Distributive law

(ii) $(a \cdot b)' = a' + b' \quad \forall a, b \in B$

$= (b + a + a') \cdot (a + b + b')$

By commutative law

$= (b + 1) \cdot (a + 1)$

By complement law

$= 1 \cdot 1$ By boundedness law

$= 1$ By identity law

$= \text{RHS}$

$$(a+b).(a'.b')$$

$$= a.(a'.b') + b.(a'.b')$$

By distributive law

$$= a.(a'.b') + b.(b'.a')$$

By commutative law

$$= (a.a').b' + (b.b').a'$$

By Associative law

$$= 0.b' + 0.a'$$

By complement law

$$= 0 + 0 \quad \text{By Boundedness law}$$

$$= 0 \quad \text{By Idempotent law}$$

$$= \text{RHS}$$

$$\Rightarrow (a+b)' = a'.b'$$

Hence proved

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DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -2 : Functions and Boolean Algebra

Lecture - 06

Today's Target

- ✓ ➤ Truth table
- ✓ ➤ Boolean Function
- ✓ ➤ Minterm
- ✓ ➤ Maxterm
- Disjunction Normal Form (D N Form)
OR
Sum of Products Form (SOP Form)

➤ The basic operations used in Boolean algebra are

(i) Logical Addition

(ii) Logical Multiplication

(iii) Complementation

0	False	OFF	LOW	OPEN	NO
1	True	ON	HIGH	CLOSED	YES

➤ Each variable in Boolean algebra has either 0 or 1 (**False or True**)

➤ Here 1 or 0 do not represent actual numbers

Logical Addition : Logical Addition is denoted by + or by OR

Truth Table for logical addition

Total input combination = 2^n

$0 + 0 = 0$ otherwise 1

INPUT		OUTPUT
A	B	$C = A + B$
1	1	1
1	0	1
0	1	1
0	0	0

Logical Multiplication : Logical Multiplication is denoted by $\bullet$ or by AND

Truth Table for logical Multiplication

$1.1 = 1$ otherwise 0

INPUT		OUTPUT
A	B	$C = A.B$
1	1	1
1	0	0
0	1	0
0	0	0

Logical Complementation : Logical Complementation is denoted by $\neg$ or by ' ($\bar{A}$, A')

Truth Table for logical Complementation

INPUT	OUTPUT
A	$B = A'$
1	0
0	1

Boolean Function (or Boolean Polynomial)

An expression obtained by the application of binary operation ($+$ or $\vee$) and ($\bullet$ or $\wedge$) and unary operation ($'$) on finite number of elements of Boolean algebra $(B, +, \bullet, ')$ or $(B, \vee, \wedge, ')$ is called a Boolean function or Boolean Polynomial.

A variable x is called a Boolean variable if it assumes values either 0 or 1.

Note:- (i) $\wedge$ or $\bullet$ $\longrightarrow$ Meet or Conjunction

(ii) $\vee$ or $+$ $\longrightarrow$ Join or Disjunction

Examples:

1. Function of one Variables

$$f(x) = x + x'$$

OR

$$f(x) = x \vee x'$$

2. Function of two variables x and y

$$f(x, y) = x \cdot y' + x' \cdot y$$

OR

$$f(x, y) = (x \wedge y') \vee (x' \wedge y)$$

3. Function of three variables x, y and z

$$f(x, y, z) = x \cdot y \cdot z + x' \cdot y \cdot z + x \cdot y' \cdot z'$$

OR

$$f(x, y, z) = (x \wedge y \wedge z) \vee (x' \wedge y \wedge z) \vee (x \wedge y' \wedge z')$$

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Minterm (OR Minimal Boolean Function)

A minterm of n -variables is a **product** of n literals in which each variable appears exactly once in either true or complemented form, but not both.

For Examples:

1. The list of all minterms of two variables x and y are

$$xy, x'y, xy', x'y'$$

3. The list of all minterms of three variables x , y and z are

$$xyz, xyz', xy'z, x'yz, x'y'z', x'y'z, x'yz', xy'z'$$

Note:

(1) n -variables can be combined to form 2^n minterm

(2) The output value of a minterm is 1

Maxterm (or Maximal Boolean Function)

A maxterm of n variables is a sum of n literals in which each variable appears exactly once in either true or complimented form, but not both.

For Examples:

1. All The maxterm of two variables x and y are

$$(2^2 = 4)$$

$$x + y, x' + y, x + y', x' + y'$$

2. All The maxterm of three variables x , y and z are

$$(2^3 = 8)$$

$$(x + y + z), (x + y + z'), (x + y' + z), (x' + y + z), (x' + y' + z'), (x' + y' + z), \\ (x' + y + z'), (x + y' + z')$$

Note:

(1) The number of maxterm in n -variables are 2^n

(2) The output value of a maxterm is 0

Note:

For Minterm:-

1 $\longrightarrow$ **Use same variable**

0 $\longrightarrow$ **Use Complement of variable**

For Maxterm:-

0 $\longrightarrow$ **Use same variable**

1 $\longrightarrow$ **Use Complement**

Truth Table:

1. For two variable x, y

x	y	Minterms(m)	Maxterms(M)
1	1	$m_1 = x \cdot y$	$M_1 = x' + y'$
1	0	$m_2 = x \cdot y'$	$M_2 = x' + y$
0	1	$m_3 = x' \cdot y$	$M_3 = x + y'$
0	0	$m_4 = x' \cdot y'$	$M_4 = x + y$

3. For three variable

x	y	z	Minterm(m)	Maxterm(M)
1	1	1	$m_1 = x.y.z$	$M_1 = x' + y' + z'$
1	1	0	$m_2 = x.y.z'$	$M_2 = x' + y' + z$
1	0	1	$m_3 = x.y'.z$	$M_3 = x' + y + z'$
1	0	0	$m_4 = x.y'.z'$	$M_4 = x' + y + z$
0	1	1	$m_5 = x'.y.z$	$M_5 = x + y' + z'$
0	1	0	$m_6 = x'.y.z'$	$M_6 = x + y' + z$
0	0	1	$m_7 = x'.y'.z$	$M_7 = x + y + z'$
0	0	0	$m_8 = x'.y'.z'$	$M_8 = x + y + z$

Boolean Function

Disjunction Normal Form

Conjunction Normal Form

Next Lecture

Disjunction Normal Form (D N Form)

OR

Sum of Product Form (SOP Form) / sum of minterm

OR

Canonical Form

A Boolean function which can be written as the sum of Minterms, is called disjunctive normal form or D N Form

Note:- As Minterm is product of variables, so D N form is also called sum of minterm or sum of product form (SOP)

STEP-1: Make a truth table for given Boolean function

$\wedge$ or $\bullet$	Conjunction Meet Product	$T \quad T \Rightarrow T$ otherwise F $1 \quad 1 \Rightarrow 1$ otherwise 0
$\vee$ or $+$	Disjunction Join Addition	$F \quad F \Rightarrow F$ otherwise T $0 \quad 0 \Rightarrow 0$ otherwise 1
$\neg$ or $'$	Complement Prime Negation	$T \longrightarrow F$ $F \longrightarrow T$ $1 \longrightarrow 0$ $0 \longrightarrow 1$

STEP-2: See the entries 1 in the last column of truth table and write corresponding minterms.

STEP-3: By adding the all minterms, we will get required D N form or SOP form

Q. 1. Express the Boolean function in to D N form $f(x, y) = x + x' \cdot y$

$$f(x, y) = x + x' \cdot y$$

x	y	x'	$x' \cdot y$	$x + x' \cdot y$	Minterm
1	1	0	0	1	$m_1 = xy$
1	0	0	0	1	$m_2 = xy'$
0	1	1	1	1	$m_3 = x'y$
0	0	1	0	0	

✓

✓

Required DNI Form

$$f(x, y) = xy + xy' + x'y$$

2nd Method

$$f(x, y) = x + x' \cdot y$$

$$f(x, y) = x \cdot 1 + x' \cdot y$$

$$f(x, y) = x(y + y') + x' \cdot y$$

$$f(x, y) = xy + xy' + x'y$$

Q. 2: Express the Boolean polynomial $f(x, y, z) = (x + y) \cdot (x + z) + y + z'$ in to disjunctive normal form.

(AKTU-2020)

x	y	z	$(x+y)$	$(x+z)$	$(x+y) \cdot (x+z)$	z'	$y+z'$	$(x+y) \cdot (x+z) + y + z'$	Minterm
1	1	1	1	1	1	0	1	1	$m_1 = xyz$
1	1	0	1	1	1	1	1	1	$m_2 = xyz'$
1	0	1	1	1	1	0	0	1	$m_3 = xy'z$
1	0	0	1	1	1	1	1	1	$m_4 = xy'z'$
0	1	1	1	1	1	0	1	1	$m_5 = x'yz$
0	1	0	1	0	0	1	1	1	$m_6 = x'yz'$
0	0	1	0	1	0	0	0	0	
0	0	0	0	0	0	1	1	1	$m_8 = x'y'z'$

Required DN Form

$$f(n, y, z) = nyz + nyz' + ny'z + ny'z' + n'yz + n'yz' + n'y'z'$$

2nd Method

$$f(n, y, z) = (n+y).(n+z) + y + z'$$

$$= n + yz + y + z'$$

$$= n.1.1 + 1.yz + 1.y.1 + 1.1.z'$$

$$= n(y+y')(z+z') + (n+n')yz + (n+n')y(z+z') + (n+n')(y+y')z'$$

$$= (ny + ny')(z + z') + nyz + n'yz + (ny + n'y)(z + z') + (n + n')(yz + y'z')$$

$$= nyz + nyz' + ny'z + ny'z' + \cancel{nyz} + \cancel{n'yz} + \cancel{nyz} + \cancel{nyz'} + \cancel{n'yz} + \cancel{n'yz'} + \cancel{nyz} + \cancel{nyz'} + \cancel{n'yz} + \cancel{n'yz'} + \cancel{nyz} + \cancel{nyz'} + \cancel{n'yz} + \cancel{n'yz'}$$

Q. 3. Let $E(x, y, z) = [(x + y)' + (x' \cdot z)]'$ be a Boolean expression. Write $E(x, y, z)$ in disjunctive normal form.

x	y	z	$x+y$	$(x+y)'$	x'	$x' \cdot z$	$(x+y)' + (x' \cdot z)$	$[(x+y)' + (x' \cdot z)]'$	Minterms
1	1	1	1	0	0	0	0	1	$m_1 = xyz$
1	1	0	1	0	0	0	0	1	$m_2 = xyz'$
1	0	1	1	0	0	0	0	1	$m_3 = xy'z$
1	0	0	1	0	0	0	0	1	$m_4 = xy'z'$
0	1	1	1	0	1	1	1	0	$m_5 = x'y z'$
0	1	0	1	0	1	0	0	1	
0	0	1	0	1	1	1	1	0	
0	0	0	0	1	1	0	0	1	

Required DN Form

$$E(n, y, z) = nyz + nyz' + ny'z + ny'z' + n'yz'$$

2nd Method

$$\begin{aligned} E(n, y, z) &= [(n+y)' + (n'.z)]' \\ &= ((n+y)')' \cdot (n'.z)' \\ &= (n+y) \cdot (n+z') \\ &= n + yz' \end{aligned}$$

$$E(n, y, z) = n + yz'$$

$$\begin{aligned} &= n(y+y')(z+z') + (n+n')yz' \\ &= (ny + ny')(z+z') + nyz' + n'yz' \\ &= nyz + nyz' + ny'z + ny'z' + \cancel{nyz'} + n'yz' \end{aligned}$$

$$E(n, y, z) = nyz + nyz' + ny'z + ny'z' + n'yz'$$

Topic : Disjunction Normal Form (D N Form)

Q.1. Represent each of the following Boolean function in disjunctive normal form.

(i) $f(x_1, x_2) = (x_1 + x_2)(x'_1 + x'_2)$

(ii) $f(x, y) = x + xy$

(iii) $f(x_1, x_2, x_3) = x_1x'_2 + x_3$

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DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -2 : Functions and Boolean Algebra

Lecture - 07

Today's Target

➤ *Conjunctive Normal Form (C N Form)*

OR

Product of Sum Form (POS Form)

➤ *Conversion of disjunctive normal form(DNF)of a boolean function to its conjunctive normal form and vice*

➤ *AKTU PYQs*

➤ *DPPs*

Boolean Function



Conjunctive Normal Form (C N Form)

OR

Product of Sum Form (POS Form)

OR

Canonical Form

A Boolean function which can be written as the product of Maxterm is called conjunctive normal form or C N Form

Note:- As Maxterm is the Sum of variables, so C N form is also called product of sum of form.

STEP-1: Make a truth table for given Boolean function

$\wedge$ or $\bullet$	Conjunction Meet Product	$T \quad T \Rightarrow T$ otherwise F $1 \quad 1 \Rightarrow 1$ otherwise 0
$\vee$ or $+$	Disjunction Join Addition	$F \quad F \Rightarrow F$ otherwise T $0 \quad 0 \Rightarrow 0$ otherwise 1
$\neg$ or $'$	Complement Prime Negation	$T \longrightarrow F$ $F \longrightarrow T$ $1 \longrightarrow 0$ $0 \longrightarrow 1$

STEP-2: See the entries 0 in the last column of truth table and write corresponding maxterms.

STEP-3: By product of all maxterm, we will get required C N form or POS form

Note:

For Minterm:-

1 → **Use same variable**

0 → **Use Complement of variable**

For Maxterm:-

0 → **Use same variable**

1 → **Use Complement**

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Q. 1:- Express the Boolean function $f(a,b,c) = a b + a' c$ as a product of maxterm.

$$f(a,b,c) = ab + a'c$$

✓ ✓

	a	b	c	ab	a'	a'c	$f(a,b,c) = ab + a'c$	Maxterm (M)
	1	1	1	1	0	0	1	
	1	1	0	1	0	0	1	
→	1	0	1	0	0	0	0	$M_1 = a' + b + c'$
→	1	0	0	0	0	0	0	$M_2 = a' + b + c$
	0	1	1	0	1	1	1	
→	0	1	0	0	1	0	0	$M_3 = a + b' + c$
	0	0	1	0	1	1	1	
→	0	0	0	0	1	0	0	$M_4 = a + b + c$

Required CN Form

$$f(a, b, c) = (a' + b + c')(a' + b + c)(a + b' + c)(a + b + c')$$

2nd Method

$$f(a, b, c) = (ab + a'c)$$

$$= (ab + a')(ab + c)$$

$$= (a' + ab)(c + ab)$$

$$= (a' + a)(a' + b)(c + a)(c + b)$$

$$= 1(a' + b)(a + c)(b + c)$$

$$= (a' + b)(a + c)(b + c)$$

✓✓

$$\textcircled{1} (a + b)' = a' \cdot b'$$

$$\textcircled{2} (ab)' = a' + b'$$

De Morgan's law

$$a + bc = (a + b) \cdot (a + c)$$

Distributive law

Q. 2:- Let $f(x, y, z) = [(x + y)' + (x' \cdot z)]'$ be a Boolean expression. Write $f(x, y, z)$ in conjunctive normal form **(AKTU-2016)**

$$f(x, y, z) = [(x + y)' + (x' \cdot z)]'$$

x	y	z	x+y	(x+y)'	x'	x'·z	(x+y)' + (x'·z)	$f(x, y, z) = [(x+y)' + (x' \cdot z)]'$
1	1	1	1	0	0	0	0	1
1	1	0	1	0	0	0	0	1
1	0	1	1	0	0	0	0	1
1	0	0	1	0	0	0	0	1
0	1	1	1	0	1	1	1	0 ✓
0	1	0	1	0	1	0	0	1
0	0	1	0	1	1	1	1	0 ✓
0	0	0	0	1	1	0	1	0 ✓

Conversion of disjunctive normal form(DNF)of a boolean function to its conjunctive normal form and vice – versa

$$DN \rightarrow CN \text{ OR } CN \rightarrow DN$$

Q.3 Convert the boolean function $f(x, y) = xy' + x'y + x'y'$ to its conjunctive normal form

$$f(x, y) = xy' + x'y + x'y'$$

Complete DNF Form for 2 variables x and y

$$f(x, y) = xy + xy' + x'y + x'y'$$

$$f'(x, y) = xy$$

$$f''(x, y) = (xy)'$$

$$f''(x, y) = x' + y'$$

Required

CNF Form

Q.4 Convert the boolean function $f(x, y, z) = (x' + y + z')(x' + y + z)(x + y' + z)$ to its disjunctive normal form.

$$f(x, y, z) = (x' + y + z')(x' + y + z)(x + y' + z)$$

complete CN Form For three variables x, y and z

$$f(x, y, z) = (x + y + z)(x + y + z')(x + y' + z)(x' + y + z)(x' + y' + z')(x' + y' + z)$$

$$f'(x, y, z) = (x + y + z)(x + y + z')(x' + y' + z')(x' + y' + z)(x + y' + z')$$

$$f''(x, y, z) = \left[(x + y + z)(x + y + z')(x' + y' + z')(x' + y' + z)(x + y' + z') \right]'$$

Q.1. Express the Boolean function in to C N form $f(x, y) = x + x' \cdot y$

Q.2. Find the product of sum expansion of each of the following

(i) $f(x, y, z) = (x + z) y$

(ii) $f(x, y, z) = xy'$

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DISCRETE STRUCTURES & THEORY OF LOGICS

(Discrete Mathematics)

UNIT - 2 : Functions and Boolean Algebra

Lecture - 08

Today's Target

- *K – Map for SOP form*
- PYQ
- DPP

1. By Karnaugh Map OR By K-Map

2. By Boolean algebra

K – Map :

The K – Map is a pictorial representation of truth table of the Boolean functions and used to minimize the Boolean functions

Note :

K – Map Matrix contain 2^n cell / square, where n is the total number of variables



1. For two variables x and y :

➤ No. of cells in K – Map = $2^2 = 4$

➤ Each cell contain a minterm

x	y'	y
	$x'y'$	$x'y$
x'	0	1
x	xy'	xy
	2	3

x	0	1
	$x'0$	$x'1$
0	0	1
1	10	11
	2	3

2. For three variables x , y and z

- No. of cells in K-Map = $2^3 = 8$
- Each cell represent a minterm

		yz			
		$y'z'$	$y'z$	yz	yz'
x	x'	$x'y'z'$ 0	$x'y'z$ 1	$x'yz$ 3	$x'yz'$ 2
	x	$xy'z'$ 4	$xy'z$ 5	xyz 7	xyz' 6

		yz			
		00	01	11	10
x	0	000 0	001 1	011 3	010 2
	1	100 4	101 5	111 7	110 6

3. For Four variables x , y , z and w

➤ No. of cells in K-Map = $2^4 = 16$

➤ Each cell represent a Minterm

$xy \backslash zw$	$z'w'$	$z'w$	zw	zw'
$x'y'$	$x'y'z'w'$ 0	$x'y'z'w$ 1	$x'y'zw$ 3	$x'y'zw'$ 2
$x'y$	$x'yz'w'$ 4	$x'yz'w$ 5	$x'yzw$ 7	$x'yzw'$ 6
xy	$xyz'w'$ 12	$xyz'w$ 13	$xyzw$ 15	$xyzw'$ 14
xy'	$xy'z'w'$ 8	$xy'z'w$ 9	$xy'zw$ 11	$xy'zw'$ 10

$xy \backslash zw$	00	01	11	10
00	0000 0	0001 1	0011 3	0010 2
01	0100 4	0101 5	0111 7	0110 6
11	1100 12	1101 13	1111 15	1110 14
10	1000 8	1001 9	1011 11	1010 10

✓ **1. Construct the K- Map**

✓ **2. Put 1 in the cell corresponding to each Minterm given in the Boolean function**

✓ **3. Making Groups of 1**

(i) Group of 1 contain 1, 2, 4, 8 Cells ✓

(ii) Diagonal groups are not allowed ✓

(iii) Overlapping allowed in groups ✓

(iv) Group should be as large as possible ✓

(v) No 1 should be as left without grouping ✓

✓ **4. Write the Minimised term for all the groups keeping in mind**

(i) Write only unchanged variable for each group

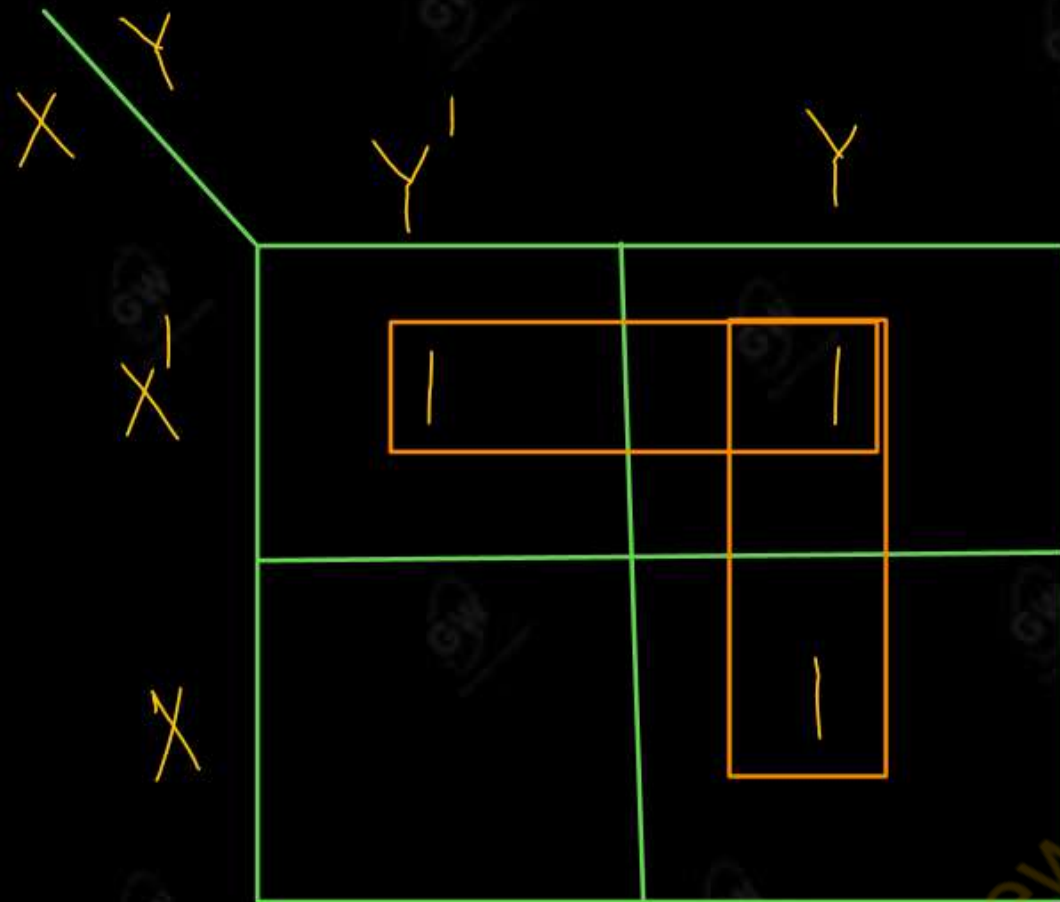
(ii) Check rows and column values for each group

5. Add all the minimised term to get required result

Q. 1. Using the K – Map to find minimal form of the Boolean functions $f(X, Y) = XY + X'Y + X'Y'$

$$= \sum(0, 1, 3)$$

K-map for two variables



Hence, the required minimal form

$$f(X, Y) = X' + Y$$

Q. 2. Using the K – Map to find minimal form of the Boolean functions $f(x, y) = xy + x'y'$

$$= \sum (0, 3)$$

K-map for two variables

		y	
		y'	y
x	x'	1	
	x		1

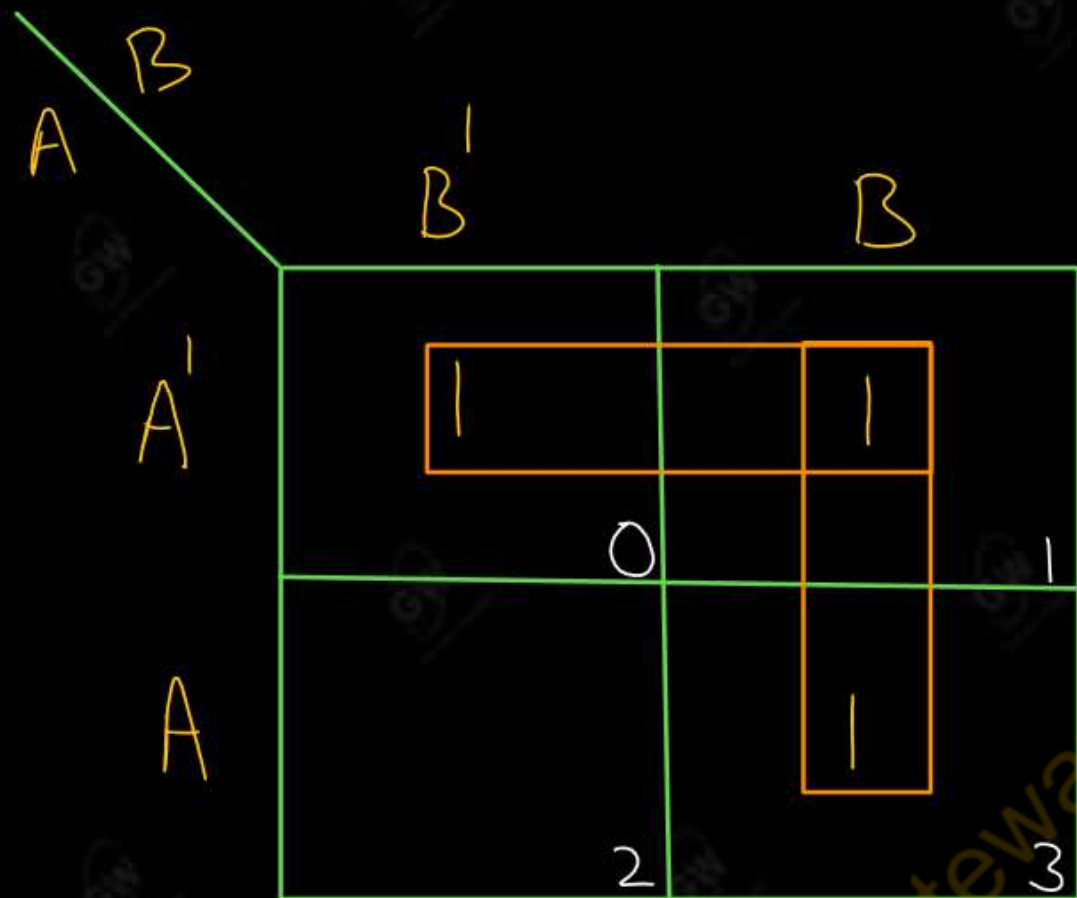
Required Minimal form

$$f(x, y) = x'y' + xy$$

Q. 3. Using the Karnaugh Map to find minimal form of the Boolean functions $f(A, B) = \Sigma(0, 1, 3)$

K-map for 2 variables

$$= A'B' + A'B + AB$$



Required Minimal form

$$f(A, B) = A' + B$$

Q. 4. Find using the K – Map a minimal form of the Boolean functions

k-map for 3 variables

$$f(x, y, z) = xyz + xyz' + xy'z + x'yz + x'y'z$$

$x \backslash yz$	$y'z'$	$y'z$	yz	yz'
x'		1	1	
x		1	1	1

Required minimal form

$$f(x, y, z) = z + xy$$

Q. 5. Find using the K – Map a minimal form of the Boolean functions

k-map for 3 variables

$$f(A, B, C) = ABC + ABC' + A'BC' + A'B'C$$

$$= \sum (1, 2, 6, 7)$$

A \ BC					
		B'C'	B'C	BC	BC'
A'	0		1		1
	1			1	1

Required minimal form

$$f(A, B, C) = AB + BC' + A'B'C$$

Q. 6. Find using the K – Map a minimal form of the Boolean functions

$$F(A, B, C) = \Sigma (0, 1, 2, 4, 5, 6)$$

K-map for 3-variables

		BC			
		$B'C'$	$B'C$	BC	BC'
A	A'	1	1		1
		0	1	3	2
	A	1	1		1
		4	5	7	6

Required minimal form

$$F(A, B, C) = B' + C'$$

Q. 7:- Use K – Map to find a minimal form

$$= \sum_m(2, 7, 8, 10, 12)$$

$$f(A, B, C, D) = A'B'CD + A'B'CD' + A'B'CD' + ABC'D' + AB'CD'$$

AB \ CD	C'D'	C'D	CD	CD'
A'B'				1
A'B			1	
AB	1			
AB'	1			1

Required minimal form

$$f(A, B, C, D) = A(C'D' + B'CD') + A'B'CD$$

Q. 8:- Simplify the Boolean function $F(A, B, C, D) = \Sigma (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11)$

v. imp

AB \ CD		CD			
		$C'D'$	$C'D$	CD	CD'
$A'B'$		1	1	1	1
		0	1	3	2
$A'B$		1	1	1	1
		4	5	7	6
AB					
		12	13	15	14
AB'		1	1	1	
		8	9	11	10

Simplified form

$$F(A, B, C, D) = A' + B'C' + B'D$$

Q. 9:- Solve the following Boolean function using K – Map

$$f(A, B, C, D) = \sum_m (0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$$

AB \ CD	CD			
	C'D'	C'D	CD	CD'
A'B'	1 0			1 2
A'B	1 4	1 5	1 7	1 6
AB		1 12	1 13	
AB'	1 8			1 10

Required minimal form

$$f(A, B, C, D) = BD + A'B + B'D'$$

Q. 10:- Solve the following Boolean functions using K – Map.

$$F(A, B, C, D) = \sum_m (m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$$

AB \ CD		CD			
		$C'D'$	$C'D$	CD	CD'
$A'B'$	1	1		1	
$A'B$	1	1		1	
AB	1	1		1	
AB'	1	1			

Required function

$$F(A, B, C, D) = C' + A'D' + BD'$$

Topic : K – Map for SOP form

- Q.1.** Using the K – Map to find minimal form of the Boolean functions $f(x, y) = x'y + xy$
- Q. 2.** Using the Karnaugh Map to find minimal form of the Boolean functions $f(A, B) = \Sigma(0, 2)$
- Q.3.** Find using the K – Map a minimal form of the Boolean functions $f(X, Y, Z) = XYZ + XYZ' + X'YZ$
- Q. 4.** Find using the K – Map a minimal form of the Boolean functions $f(A, B, C) = \Sigma_m(0, 2, 3, 4, 7)$
- Q. 5.** Use K – Map to find a minimal form $f(x, y, z, w) = xyzw + xyzw' + xy'zw' + x'y'zw + x'y'zw'$
- Q.6** Solve $E(x, y, z, t) = \Sigma(0, 2, 6, 8, 10, 12, 14, 15)$ using K – Map.
- Q.7.** Solve the following Boolean functions using K – Map $F(A, B, C, D) = \Sigma(0, 2, 5, 7, 8, 10, 13, 15)$

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Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS

(Discrete Mathematics)

UNIT - 2 : Functions and Boolean Algebra

Lecture - 09

Today's Target

- *K – Map for POS form*
- *Simplification of Boolean function by Boolean algebra*
- PYQ
- DPP

K - Map

For SOP form

OR

(Sum of Minterm)

OR

(DN Form)

For POS form

OR

(Product of Maxterm)

OR

(CN Form)

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K – Map for POS form

1. For two variables x and y :

➤ No. of cells in K – Map = 4

➤ Each cell contain a Maxterm

x		y		y'	
		y	y'		
x	$x+y$	$x+y'$	0	1	
	$x'+y$	$x'+y'$	2	3	
x'					

x		y		y'	
		y	y'		
x	00	01	0	1	
	10	11	2	3	
x'					

2. For three variables x , y and z

No. of cells in K-Map = $2^3 = 8$

Each cell represent a Maxterm

$x \backslash yz$		yz	$y+z'$	$y'+z'$	$y'+z$
x	x				
	x'				
		0	1	3	2
		4	5	7	6

$x \backslash yz$		00	01	11	10
x	0				
	1				
		0	1	3	2
		4	5	7	6

3. For Four variables x, y, z and w

No. of cells in K-Map = $2^4 = 16$

Each cell represent a Maxterm

xy zw

	$z+w$	$z+w'$	$z'+w'$	$z'+w$
$x+y$	0	1	3	2
$x+y'$	4	5	7	6
$x'+y'$	12	13	15	14
$x'+y$	8	9	11	10

xy zw

	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

1. Construct Karnaugh Map

2. Put 0 in the cell corresponding to each *Maxterm* given in the Boolean function

3. Making Groups of 0

- (i) Group of 0 contain 1, 2, 4, 8 Cells
- (ii) Diagonal groups are not allowed
- (iii) Overlapping allowed in groups
- (iv) Group should be as large as possible
- (v) No 0 should be left without grouping

4. Write the Minimised term for all the groups keeping in mind

- (i) Write only unchanged variable for each group
- (ii) Check rows and column values for each group

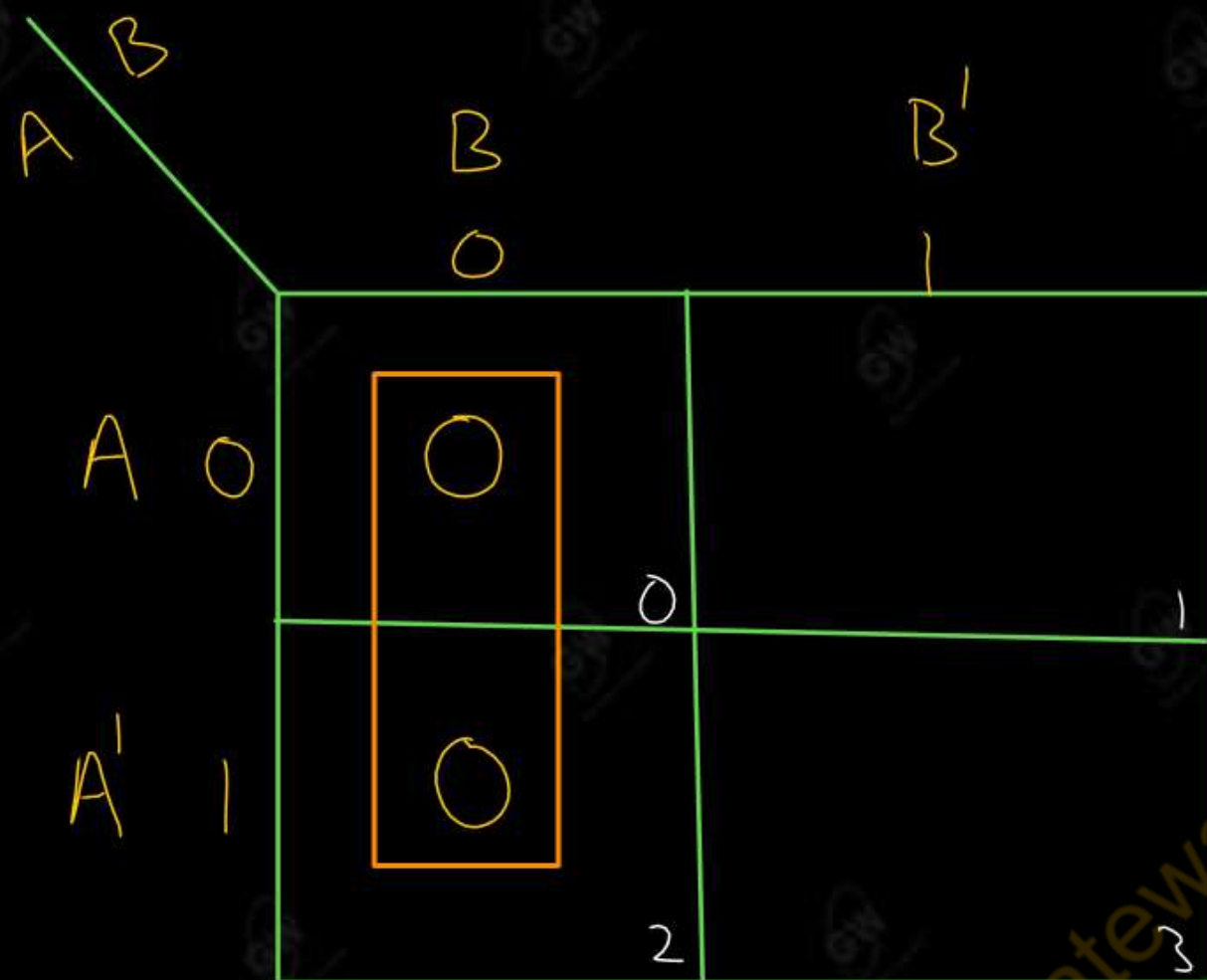
5. Add all the minimised term to get required result

Q. 1:- Minimise the Boolean expression $f(A, B) = \prod M(0, 2)$

$$= (A + B)(A' + B)$$

Required minimal form

$$f(A, B) = B$$



Q. 2:- Draw the K-map and simplify the Boolean expression $F(A, B, C) = \prod M(0, 1, 4, 5)$

$$= (A + B + C)(A + B + C')(A' + B + C)(A' + B + C')$$

BC					
		$B+C$ 00	$B+C'$ 01	$B'+C'$ 11	$B'+C$ 10
A	0	0	0		
	1	0	0		
		4	5	7	6

Required simplified form

$$F(A, B, C) = B$$

Q. 3:- Draw the K – map and simplify the Boolean function

$$F(A, B, C, D) = \prod M(1, 2, 5, 6, 8, 9, 15)$$

AB \ CD				
	C + D 00	C + D' 01	C' + D' 11	C' + D 10
00 A + B		0	1	3
01 A + B'		4	5	7
11 A' + B'		12	13	15
10 A' + B	8	9	11	14

Simplified Boolean function

$$F(A, B, C, D) = (A + C + D')$$

$$(A + C' + D)(A' + B + C)$$

$$(A' + B' + C' + D')$$

Q. 4:- Simplify using K-map to obtain a minimum POS expression:

$$F(A, B, C, D) = (A' + B' + C + D)(A + B' + C + D)(A + B + C' + D')(A' + B + C' + D')(A + B + C' + D)$$

AB \ CD					
		C + D 00	C + D' 01	C' + D' 11	C' + D 10
A + B 00	A + B'			0	0
		0			
		0			
				0	

Simplified Boolean function

$$F(A, B, C, D) = (B' + C + D)$$

$$(A + B + C')(B + C' + D')$$

Q.5 Simplify the Boolean expressions

(a) $C(B + C)(A + B + C)$

(b) $XY + X'Z + YZ$

(c) $A + B(A + B) + A(A' + B)$

(a) $C(B + C)(A + B + C)$

$= (CB + CC)(A + B + C)$

$= (CB + C)(A + B + C) \left\{ \because CC = C \right\}$

$= C(A + B + C) \left\{ \because a + ac = a \right\}$

$= C(A + B + C)$

$= CA + CB + C \left\{ a + ac = a \right\}$

$= C(A + C)$

$= C$

(b) $XY + X'Z + YZ$

$= XY + X'Z + YZ(X + X')$

$\left\{ X + X' = 1 \right\}$

$= \underline{XY} + X'Z + \underline{XYZ} + X'YZ$

$$= XY + XYZ + X'Z + X'ZY$$

$$= XY(1+Z) + X'Z(1+Y)$$

$$= XY + X'Z \quad \left\{ \because a+1=1 \right\}$$

$$(C) A + B(A+B) + A(A'+B)$$

$$= A + AB + BB + AA' + AB$$

$$= A + AB + AB + B + 0 \quad \left\{ \begin{array}{l} \because a \cdot a' = 0 \\ a \cdot a = a \end{array} \right.$$

$$= A + AB + B \quad \left\{ \because a+a=a \right\}$$

$$= A + B \quad \left\{ a+a=a \right\}$$

DPP- 9

Topic : K – Map for POS form

Q.1. Draw the K – map and simplify the Boolean function $F(A, B, C, D) = \prod M(1, 2, 4, 6, 8, 9)$

Q.2. Draw the K – map and simplify the Boolean function $F(A, B, C, D) = \prod M(3, 4, 6, 7, 11, 12, 13, 14, 15)$

Q. 3. Simplify the following using Boolean algebra

(a) $X = (A + B + AB)(A + C)$

(b) $A = XY + (XZ)' + XY'Z (XY + Z)$

(c) $A = XY + XYZ + X'Y + XY'Z$

(d) $A = X [Y + Z (XY + XZ)']$

(e) $A = XY + X'YZ' + YZ$

(f) $A = (XY' + Z)(X + Y')Z$

(g) $A = XY' + Z + (X' + Y) Z'$

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Thank You